

Additional file 1: Supplementary tables, notes and figures

Ensembles Across Model Families Outperform Any Single Architecture:
A Single-Configuration, Leakage-Controlled Benchmark of 31 Molecular Property Models
Across 44 Datasets

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This file contains the supplementary tables (S1–S15), notes (S1–S6) and figures (S1–S3) referenced in the main text. Notes S1, S2, S5 and S6 hold the detail that the revised main text summarizes; Note S3 reports two sensitivity analyses of the family comparison; Note S4 is the dataset-property meta-analysis. Tables S1–S7 and S12 are generated from the deposited benchmark artifacts by `portable_colab_qsar_bundle/render_manuscript_assets.py` and `render_latex_tables.py`; Table S8 by `python -m qsarena.reliability_tables` from the reliability-study artifacts; Tables S9–S11, Figures S1–S3 and the text of Note S4 by `qsarena.meta_analysis`, which the same render script runs; and Tables S13–S14 by `reanalysis_coverage_posthoc.py`. They carry the same values as the machine-readable CSV and Markdown files in the repository.

All results derive from a single split and a single seed per dataset; as set out in the Limitations section of the main text, individual entries are therefore provisional and close comparisons between models should not be treated as established.

Table S1: Per-dataset winning model under the ranking metric, with the corresponding cross-validation-selected model and its relative gap to the dataset best (%).

dataset	suite	task - kind	family	model	analysis - analysis_metric_value	cv - selected - model	cv - selected - family	cv_selected_gap_to_best	
tdc_ames	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_roc_auc	0.883	Extra trees	Conventional ML	1.386
tdc_bbb_martins	TDC	classification	Conventional ML	Random forest	test_roc_auc	0.925	XGBoost (ADMETboost features)	Conventional ML	0.963
tdc_bioavailability_ma	TDC	classification	Conventional ML	CatBoost	test_roc_auc	0.777	XGBoost (ADMETboost features)	Conventional ML	7.129
tdc_carcinogens_lagunin	TDC	classification	Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=3)	test_roc_auc	0.929	XGBoost (ADMETboost features)	Conventional ML	2.237
tdc_clintox	TDC	classification	CFA combinatorial fusion	CFA (Combinatorial Fusion)	test_roc_auc	0.974	AdaBoost	Conventional ML	8.986
tdc_cyp1a2_veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse OOF error))	test_roc_auc	0.974	SVC	Conventional ML	1.196
tdc_cyp2c19_veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_roc_auc	0.938	XGBoost	Conventional ML	1.338
tdc_cyp2c9_substrate_carbonmangels	TDC	classification	Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=3)	test_auprc	0.438	SVC	Conventional ML	31.368
tdc_cyp2c9_veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_auprc	0.827	AdaBoost	Conventional ML	14.709
tdc_cyp2d6_substrate_carbonmangels	TDC	classification	Chemprop v2 GNN	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	test_auprc	0.693	ChemML MLP (PyTorch)	Deep tabular NN (ChemML MLP)	26.129
tdc_cyp2d6_veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_auprc	0.763	AdaBoost	Conventional ML	19.417
tdc_cyp3a4_substrate_carbonmangels	TDC	classification	Conventional ML	XGBoost (ADMETboost features)	test_auprc	0.729	XGBoost (ADMETboost features)	Conventional ML	0.000
tdc_cyp3a4_veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_auprc	0.905	AdaBoost	Conventional ML	11.894

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dataset	suite	task - kind	family	model	analysis_ metric	analysis_metric_value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
tdc_dili	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ roc_ auc	0.933	XGBoost (ADMETboost features)	Conventional ML	1.072
tdc_herg	TDC	classification	Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=3)	test_ roc_ auc	0.857	XGBoost (ADMETboost features)	Conventional ML	7.662
tdc_herg_karim	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse OOF error))	test_ roc_ auc	0.907	XGBoost	Conventional ML	0.494
tdc_hia_hou	TDC	classification	Conventional ML	XGBoost (ADMETboost features)	test_ roc_ auc	0.994	XGBoost (ADMETboost features)	Conventional ML	0.000
tdc_pampa_ ncats	TDC	classification	Uni-Mol (3D pretrained)	Uni-Mol V1	test_ roc_ auc	0.763	Random forest	Conventional ML	10.598
tdc_pgp_ broccatelli	TDC	classification	Uni-Mol (3D pretrained)	Uni-Mol V2 (84m)	test_ roc_ auc	0.933	Random forest	Conventional ML	3.065
tdc_skin_ reaction	TDC	classification	Conventional ML	XGBoost (ADMETboost features)	test_ roc_ auc	0.667	XGBoost (ADMETboost features)	Conventional ML	0.000
tdc_tox21	TDC	classification	Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=3)	test_ roc_ auc	0.597	Extra trees	Conventional ML	8.142
tdc_toxcast	TDC	classification	Conventional ML	HistGradientBoosting	test_ roc_ auc	0.716	XGBoost (ADMETboost features)	Conventional ML	1.662
chemml_cep_ homo	ChemML	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.090	TabPFNRegressor	TabPFN (tabular foundation)	13.888
chemml_ organic_density	ChemML	regression	TabPFN (tabular foundation)	TabPFNRegressor	test_ rmse	0.005	TabPFNRegressor	TabPFN (tabular foundation)	0.000
esol_delaney	Molecule Net	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse OOF error))	test_ rmse	0.622	TabPFNRegressor	TabPFN (tabular foundation)	8.211
freesolv_sampl	Molecule Net	regression	TabPFN (tabular foundation)	TabPFNRegressor	test_ rmse	0.933	XGBoost (ADMETboost features)	Conventional ML	121.064
lipophilicity	Molecule Net	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.507	TabPFNRegressor	TabPFN (tabular foundation)	17.351
poduam_pod_ nc_std	PODUAM	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.694	XGBoost (ADMETboost features)	Conventional ML	5.048

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dataset	suite	task - kind	family	model	analysis_ metric	analysis_metric_value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
poduam_pod_ rd_std	PODUAM	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.556	XGBoost (ADMETboost features)	Conventional ML	1.824
polaris_adme_ fang_hppb_1	Polaris	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.422	XGBoost (ADMETboost features)	Conventional ML	9.517
polaris_adme_ fang_perm_1	Polaris	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.379	XGBoost (ADMETboost features)	Conventional ML	16.319
polaris_adme_ fang_rclint_1	Polaris	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.484	XGBoost	Conventional ML	11.804
polaris_adme_ fang_rppb_1	Polaris	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.478	XGBoost (ADMETboost features)	Conventional ML	6.138
polaris_adme_ fang_solu_1	Polaris	regression	Uni-Mol (3D pretrained)	Uni-Mol V2 (84m)	test_ rmse	0.551	XGBoost (ADMETboost features)	Conventional ML	7.782
tdc_caco2_wang	TDC	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.338	XGBoost (ADMETboost features)	Conventional ML	11.813
tdc_clearance_ hepatocyte_az	TDC	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	44.113	XGBoost (ADMETboost features)	Conventional ML	6.388
tdc_clearance_ microsome_az	TDC	regression	Uni-Mol (3D pretrained)	Uni-Mol V2 (84m)	test_ rmse	33.507	XGBoost (ADMETboost features)	Conventional ML	13.113
tdc_half_life_ obach	TDC	regression	Uni-Mol (3D pretrained)	Uni-Mol V1	test_ rmse	19.128	XGBoost (ADMETboost features)	Conventional ML	77.574
tdc_ hydrationfreeenergy_ freesolv	TDC	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse OOF error))	test_ rmse	1.075	XGBoost (ADMETboost features)	Conventional ML	5.821
tdc_ld50_zhu	TDC	regression	TabPFN (tabular foundation)	TabPFNRegressor	test_ rmse	0.806	XGBoost	Conventional ML	4.826
tdc_ lipophilicity_ astrazeneca	TDC	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.551	TabPFNRegressor	TabPFN (tabular foundation)	8.398
tdc_ppbr_az	TDC	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	10.768	TabPFNRegressor	TabPFN (tabular foundation)	21.078
tdc_solubility_ aqsoldb	TDC	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.955	XGBoost (ADMETboost features)	Conventional ML	4.572

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dataset	suite	task_ kind	family	model	analysis_ metric	analysis_ metric_ value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
tdc_vdss_ lombardo	TDC	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	4.670	XGBoost (ADMETboost features)	Conventional ML	69.878

Table S2: Staged component ablation: datasets whose best achievable primary metric improved when each pipeline stage was added, computed from the existing model results.

stage_order	stage	datasets_evaluated	datasets_improved_vs_previous	improvement_fraction
1	Conventional ML only	44	0	0.000
2	+ Map Light classic fea- tures	44	9	0.205
3	+ neu- ral/ deep back- ends	44	26	0.591
4	+ CFA fusion	44	6	0.136
5	Full pipeline incl. ensem- bles	44	22	0.500

Table S3: Feature-family selection summary across all datasets, sorted by per-feature enrichment relative to a uniform-selection baseline.

Feature family	Datasets selected	Available (sum)	Selected (sum)	% of selected	Enrichment vs uniform
rdkit	40	8,261	1,362	11.22	5.16
erg	38	12,552	1,146	9.44	2.86
avalon	43	44,384	3,603	29.69	2.54
maccs	38	5,962	224	1.85	1.18
maplight	41	30,070	854	7.04	0.89
fcfp6	43	42,500	998	8.22	0.73
atom_pair	41	39,651	830	6.84	0.66
rdk_path	40	45,036	896	7.38	0.62
topological_torsion	40	23,653	421	3.47	0.56
ecfp6	37	43,299	731	6.02	0.53
layered	40	43,642	674	5.55	0.48
morgan	38	40,827	398	3.28	0.31

Table S4: Model coverage: datasets attempted and datasets yielding a valid metric, per model.

model	datasets_attempted	datasets_valid
Uni-Mol V2 (310m)	8	0
Uni-Mol V2 (164m)	4	0
Uni-Mol V2 (84m)	44	17
TabPFNClassifier	22	19
ElasticNetCV	22	20
LogisticRegression	22	22
MapLight CatBoost (Strict Parity)	22	22
TabPFNRegressor	22	22
SVC	22	22
SVR	22	22
Voting Classifier (KNN, SVM)	22	22
Voting Regressor (KNN, SVM)	22	22
Tabular CNN	22	22
Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	44	38
Chemprop v2 (AttentiveFP, ensemble=3)	44	38
Chemprop v2 (D-MPNN + Selected descriptors, ensemble=3)	44	42
Chemprop v2 (D-MPNN, ensemble=3)	44	42
Chemprop v2 (CMPNN, ensemble=3)	44	42
HistGradientBoosting	44	44
Ensemble (OOF Stacking (RidgeCV, 5-fold))	44	44
ChemML MLP (PyTorch)	44	44
CatBoost	44	44
CFA (Combinatorial Fusion)	44	44
ChemML MLP (TensorFlow)	44	44
AdaBoost	44	44
Extra trees	44	44
Ensemble (Weighted average (inverse OOF error))	44	44
Random forest	44	44
MapLight + GNN (CatBoost, Strict Parity)	44	44
Uni-Mol V1	44	44
Tabular MLP	44	44
XGBoost	44	44
XGBoost (ADMETboost features)	44	44

Table S5: Run-to-run comparison: the canonical NSF ACCESS Jetstream2 A100 benchmark against the earlier consumer-GPU (RTX 4060) run, analysed identically. “Same split” marks datasets where both runs used the identical held-out partition; the remainder were re-split to scaffold splits in the A100 run and are therefore not directly comparable. Change is relative and metric-direction aware (positive favours the A100 run).

Dataset	Task	Same split	A100 best model	A100 value	RTX 4060 best model	RTX 4060 value	Change (%)
tdc_ames	classification	True	TabPFNClassifier	0.876	XGBoost	0.875	0.166
tdc_bbb_martins	classification	True	Random forest	0.925	Random forest	0.932	-0.788
tdc_bioavailability_ma	classification	True	CatBoost	0.777	CatBoost	0.777	0.000
tdc_cyp1a2_veith	classification	True	XGBoost	0.969	Uni-Mol V1	0.969	-0.049
tdc_cyp2c19_veith	classification	True	XGBoost	0.926	Uni-Mol V1	0.921	0.551
tdc_cyp2c9_substrate_carbonmangels	classification	True	Chemprop v2 (AttentiveFP, ensemble=3)	0.438	Uni-Mol V1	0.454	-3.439
tdc_cyp2c9_veith	classification	True	XGBoost	0.796	HistGradientBoosting	0.792	0.541
tdc_cyp2d6_substrate_carbonmangels	classification	True	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	0.693	Random forest	0.665	4.143
tdc_cyp2d6_veith	classification	True	XGBoost	0.729	XGBoost	0.722	0.904
tdc_cyp3a4_substrate_carbonmangels	classification	True	XGBoost (ADMETboost features)	0.729	LogisticRegression	0.717	1.759
tdc_cyp3a4_veith	classification	True	XGBoost (ADMETboost features)	0.883	XGBoost	0.883	0.002
tdc_dili	classification	True	XGBoost (ADMETboost features)	0.923	Uni-Mol V1	0.914	0.903
tdc_herg	classification	True	Chemprop v2 (AttentiveFP, ensemble=3)	0.857	AdaBoost	0.861	-0.411
tdc_herg_karim	classification	True	XGBoost	0.902	Uni-Mol V1	0.893	1.055
tdc_hia_hou	classification	True	XGBoost (ADMETboost features)	0.994	CatBoost	0.989	0.499
tdc_pgp_broccatelli	classification	True	Uni-Mol V2 (84m)	0.933	TabPFNClassifier	0.927	0.625
chemml_cep_homo	regression	True	TabPFNRegressor	0.102	TabPFNRegressor	0.086	-19.231
chemml_organic_density	regression	True	TabPFNRegressor	0.005	TabPFNRegressor	0.005	5.907
esol_delaney	regression	True	MapLight CatBoost (Strict Parity)	0.645	TabPFNRegressor	0.612	-5.269
freesolv_sampl	regression	True	TabPFNRegressor	0.933	Chemprop v2 (AttentiveFP, ensemble=1)	1.080	13.562
lipophilicity	regression	True	Chemprop v2 (D-MPNN, ensemble=3)	0.551	Uni-Mol V1	0.592	6.932
poduam_pod_nc_std	regression	True	Random forest	0.720	Extra trees	0.701	-2.720
poduam_pod_rd_std	regression	True	XGBoost (ADMETboost features)	0.567	XGBoost	0.551	-2.872
polaris_adme_fang_hppb_1	regression	True	Chemprop v2 (D-MPNN, ensemble=3)	0.444	MapLight + GNN (CatBoost, Strict Parity)	0.449	1.072
polaris_adme_fang_perm_1	regression	True	Uni-Mol V1	0.399	Uni-Mol V1	0.399	-0.040
polaris_adme_fang_rclint_1	regression	True	Chemprop v2 (D-MPNN, ensemble=3)	0.517	Uni-Mol V1	0.512	-0.968
polaris_adme_fang_rppb_1	regression	True	MapLight + GNN (CatBoost, Strict Parity)	0.493	MapLight + GNN (CatBoost, Strict Parity)	0.494	0.170
polaris_adme_fang_solu_1	regression	True	Uni-Mol V2 (84m)	0.551	Uni-Mol V1	0.575	4.202
tdc_caco2_wang	regression	True	MapLight CatBoost (Strict Parity)	0.350	MapLight CatBoost (Strict Parity)	0.350	0.000
tdc_clearance_hepatocyte_az	regression	True	MapLight + GNN (CatBoost, Strict Parity)	44.113	MapLight + GNN (CatBoost, Strict Parity)	43.976	-0.311
tdc_clearance_microsome_az	regression	True	Uni-Mol V2 (84m)	33.507	Uni-Mol V1	35.888	6.634

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Dataset	Task	Same split	A100 best model	A100 value	RTX 4060 best model	RTX 4060 value	Change (%)
tdc_half_life_obach	regression	True	Uni-Mol V1	19.128	Uni-Mol V1	17.222	-11.067
tdc_ld50_zhu	regression	True	TabPFNRegressor	0.806	HistGradientBoosting	0.839	3.931
tdc_lipophilicity_astrazeneca	regression	True	Chemprop v2 (D-MPNN, ensemble=3)	0.577	Uni-Mol V1	0.587	1.698
tdc_ppbr_az	regression	True	MapLight + GNN (CatBoost, Strict Parity)	11.356	MapLight + GNN (CatBoost, Strict Parity)	11.444	0.762
tdc_solubility_aqsolddb	regression	True	TabPFNRegressor	0.961	Uni-Mol V1	0.989	2.807
tdc_vdss_lombardo	regression	True	MapLight + GNN (CatBoost, Strict Parity)	4.670	MapLight + GNN (CatBoost, Strict Parity)	4.668	-0.035
tdc_carcinogens_lagunin	classification	False	Chemprop v2 (AttentiveFP, ensemble=3)	0.929	AdaBoost	0.863	7.656
tdc_clintox	classification	False	Random forest	0.956	CatBoost	0.949	0.788
tdc_pampa_ncats	classification	False	Uni-Mol V1	0.763	Extra trees	0.805	-5.206
tdc_skin_reaction	classification	False	XGBoost (ADMETboost features)	0.667	TabPFNClassifier	0.747	-10.737
tdc_tox21	classification	False	Chemprop v2 (AttentiveFP, ensemble=3)	0.597	XGBoost	0.812	-26.451
tdc_toxcast	classification	False	HistGradientBoosting	0.716	CatBoost	0.791	-9.562
tdc_hydrationfreeenergy_freesolv	regression	False	TabPFNRegressor	1.116	TabPFNRegressor	0.592	-88.600

Table S6: Dataset catalog. “Ranking metric” is the metric used for cross-dataset model ranking (RMSE for regression; the dataset’s designated primary classification metric otherwise); leaderboard comparisons in Table S7 use the leaderboard’s own metric, which differs for several TDC regression tasks.

Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking metric	Best model	Best value	QSARena			Best published	Best published model
											Leaderboard metric	(lb metric)	Est. rank		
tdc_ames	TDC	classification	7,278	5,821	1,457	predefined	raw	roc_auc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.883	ROC_-AUC	0.883	2	0.912	QW-MTL
tdc_bbb_martins	TDC	classification	2,030	1,624	406	predefined	raw	roc_auc	Random forest	0.925	ROC_-AUC	0.925	2	0.941	MolGPS (3B)
tdc_bioavailability_ma	TDC	classification	640	512	128	predefined	raw	roc_auc	CatBoost	0.777	ROC_-AUC	0.777	1	0.748	MaxQsaring
tdc_carcinogens_lagunin	TDC	classification	280	223	57	scaffold	raw	roc_auc	Chemprop v2 (AttentiveFP, ensemble=3)	0.929	ROC_-AUC	0.929	1	0.848	FATE-Tox (MTL)
tdc_clintox	TDC	classification	1,478	1,180	298	scaffold	raw	roc_auc	CFA (Combinatorial Fusion)	0.974	ROC_-AUC	0.974	3	0.996	PrismNet
tdc_cyp1a2_veith	TDC	classification	12,579	10,061	2,518	scaffold	raw	roc_auc	Ensemble (Weighted average (inverse OOF error))	0.974					
tdc_cyp2c19_veith	TDC	classification	12,665	10,131	2,534	scaffold	raw	roc_auc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.938					
tdc_cyp2c9_substrate_carbonmangels	TDC	classification	669	534	135	predefined	raw	auprc	Chemprop v2 (AttentiveFP, ensemble=3)	0.438	AUPRC	0.438	3	0.450	MaxQsaring
tdc_cyp2c9_veith	TDC	classification	12,092	9,673	2,419	predefined	raw	auprc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.827	AUPRC	0.827	3	0.877	MaxQsaring
tdc_cyp2d6_substrate_carbonmangels	TDC	classification	667	532	135	predefined	raw	auprc	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	0.693	AUPRC	0.693	4	0.766	MaxQsaring
tdc_cyp2d6_veith	TDC	classification	13,130	10,504	2,626	predefined	raw	auprc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.763	AUPRC	0.763	2	0.811	MaxQsaring
tdc_cyp3a4_substrate_carbonmangels	TDC	classification	670	535	135	predefined	raw	auprc	XGBoost (ADMETboost features)	0.729	ROC_-AUC	0.665	6	0.692	MolE

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Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking metric	Best model	Best value	Leaderboard metric	QSArena (lb metric)	Est. rank	Best published	Best published model
tdc_cyp3a4_veith	TDC	classification	12,328	9,861	2,467	predefined	raw	auprc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.905	AUPRC	0.905	2	0.923	MaxQsaring
tdc_dili	TDC	classification	475	379	96	predefined	raw	roc_auc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.933	ROC_-AUC	0.933	4	0.945	Meta-model (NIST)
tdc_herg	TDC	classification	655	523	132	predefined	raw	roc_auc	Chemprop v2 (AttentiveFP, ensemble=3)	0.857	ROC_-AUC	0.857	3	0.880	MaxQsaring
tdc_herg_karim	TDC	classification	13,445	10,755	2,690	scaffold	raw	roc_auc	Ensemble (Weighted average (inverse OOF error))	0.907					
tdc_hia_hou	TDC	classification	578	461	117	predefined	raw	roc_auc	XGBoost (ADMETboost features)	0.994	ROC_-AUC	0.994	2	0.994	MiniMol (GINE)
tdc_pampa_ncats	TDC	classification	2,034	1,626	408	scaffold	raw	roc_auc	Uni-Mol V1	0.763					
tdc_pgp_broccatelli	TDC	classification	1,218	973	245	predefined	raw	roc_auc	Uni-Mol V2 (84m)	0.933	ROC_-AUC	0.933	4	0.994	MiniMol (GINE)
tdc_skin_reaction	TDC	classification	404	289	115	scaffold	raw	roc_auc	XGBoost (ADMETboost features)	0.667	ROC_-AUC	0.667	>10	0.741	FATE-Tox (MTL)
tdc_tox21	TDC	classification	7,258	5,797	1,461	scaffold	raw	roc_auc	Chemprop v2 (AttentiveFP, ensemble=3)	0.597	ROC_-AUC	0.597	>10	0.867	PrismNet
tdc_toxcast	TDC	classification	1,731	1,357	374	scaffold	raw	roc_auc	HistGradient Boosting	0.716	ROC_-AUC	0.716	1	0.714	PrismNet
chemml_cep_homo	ChemML	regression	500	400	100	target_-	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.090					
chemml_organic_density	ChemML	regression	500	400	100	quartiles target_-	log10	rmse	TabPFNRegressor	0.005					
esol_delaney	MoleculeNet	regression	1,128	874	254	quartiles scaffold	raw	rmse	Ensemble (Weighted average (inverse OOF error))	0.622	RMSE	0.622	7	0.558	GCN
freesolv_sampler	MoleculeNet	regression	642	513	129	random	raw	rmse	TabPFNRegressor	0.933					
lipophilicity	MoleculeNet	regression	4,200	3,357	843	scaffold	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.507	RMSE	0.507	1	0.549	GCN

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Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking metric	Best model	Best value	Leaderboard metric	QSArena (lb metric)	Est. rank	Best published	Best published model
poduam_pod_nc_std	PODUAM	regression	1,842	1,473	369	target_- raw	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.694	RMSE	0.694	2	0.550	PODUAM BNN (PODnc)
poduam_pod_rd_std	PODUAM	regression	2,355	1,884	471	quartiles target_- raw	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.556	RMSE	0.556	2	0.410	PODUAM BNN (PODr)
polaris_adme_fang_hppb_1	Polaris	regression	1,808	1,446	362	quartiles predefined	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.422	MSE	0.178	4	0.143	LateFusion-k5
polaris_adme_fang_perm_1	Polaris	regression	2,642	2,113	529	predefined	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.379	MSE	0.144	4	0.113	1B_MPNN_MolGPS-ens_Large Mix
polaris_adme_fang_rclint_1	Polaris	regression	3,054	2,443	611	predefined	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.484	MSE	0.234	5	0.204	LateFusion-k5
polaris_adme_fang_rppb_1	Polaris	regression	885	708	177	predefined	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.478	MSE	0.228	1	0.230	1B_MPNN_LargeMix-and-Phenomics
polaris_adme_fang_solu_1	Polaris	regression	2,173	1,738	435	predefined	raw	rmse	Uni-Mol V2 (84m)	0.551	MSE	0.304	7	0.222	1B_MPNN_MolGPS-ens_Large Mix
tdc_caco2_wang	TDC	regression	910	728	182	predefined	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.338	MAE	0.269	3	0.256	CaliciBoost
tdc_clearance_hepatocyte_az	TDC	regression	1,213	970	243	predefined	raw	rmse	MapLight + GNN (CatBoost, Strict Parity)	44.113	SPEARMAN	0.516	3	0.633	CFA
tdc_clearance_microsome_az	TDC	regression	1,102	881	221	predefined	raw	rmse	Uni-Mol V2 (84m)	33.507	SPEARMAN	0.678	1	0.652	MapLight + GNN
tdc_half_life_obach	TDC	regression	667	532	135	predefined	raw	rmse	Uni-Mol V1	19.128	SPEARMAN	0.597	2	0.649	CFA
tdc_hydrationfreeenergy_freesolv	TDC	regression	642	490	152	scaffold raw	raw	rmse	Ensemble (Weighted average (inverse OOF error))	1.075	RMSE	1.075	7	0.654	PrismNet

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Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking metric	Best model	Best value	QSARena			Best published	Best published model
											Leaderboard metric	(lb metric)	Est. rank		
tdc_ld50_zhu	TDC	regression	7,385	5,907	1,478	predefined	draw	rmse	TabPFNRegressor	0.806	MAE	0.552	2	0.292	Base Boosting KyQVZ6b2
tdc_lipophilicity_astrazeneca	TDC	regression	4,200	3,360	840	predefined	draw	rmse	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.551	MAE	0.412	3	0.406	MiniMol
tdc_ppbr_az	TDC	regression	2,790	2,231	559	predefined	draw	rmse	Ensemble (OOF Stacking (RidgeCV, 5-fold))	10.768	MAE	7.118	2	0.679	Gradient Boost
tdc_solubility_aqsoldb	TDC	regression	9,980	7,985	1,995	predefined	draw	rmse	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.955	MAE	0.676	2	0.557	MiniMol
tdc_vdss_lombardo	TDC	regression	1,130	904	226	predefined	draw	rmse	MapLight + GNN (CatBoost, Strict Parity)	4.670	SPEARMAN	0.680	5	0.942	MapLight + GNN

Table S7: Per-dataset leaderboard comparison, sorted by estimated rank. "References (n)" is the number of published values available for that dataset and metric; ranks from sparse reference sets are correspondingly uncertain. The **tdc_ppbr_az** top-1 reference (MAE 0.679) is inconsistent in scale with the rest of that dataset's references (top-10 cutoff 7.914) and its top-1 gap should be disregarded.

Dataset	Metric	QSARena best model (test-selected)	QSARena value	Reference top-1	Reference top-10 cutoff	References (n)	Est. rank	CV-selected model	CV-selected value	CV-selected est. rank
lipophilicity	RMSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.507	0.549	0.610	27	1	TabPFNRegressor	0.595	8
polaris_adme_fang_rppb_1	MSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.228	0.230	0.634	10	1	XGBoost (ADMETboost features)	0.257	3
tdc_bioavailability_ma	ROC_- AUC	CatBoost	0.777	0.748	0.640	8	1	XGBoost (ADMETboost features)	0.721	2
tdc_carcinogens_lagunin	ROC_- AUC	Chemprop v2 (AttentiveFP, ensemble=3)	0.929	0.848	0.795	20	1	XGBoost (ADMETboost features)	0.908	1
tdc_clearance_microsome_az	SPEARMA	AutoMol V2 (84m)	0.678	0.652	0.599	17	1	XGBoost (ADMETboost features)	0.549	>10
tdc_toxcast	ROC_- AUC	HistGradientBoosting	0.716	0.714	0.714	2	1	XGBoost (ADMETboost features)	0.704	3
poduam_pod_nc_std	RMSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.694	0.550	0.730	2	2	XGBoost (ADMETboost features)	0.729	2
poduam_pod_rd_std	RMSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.556	0.410	0.630	2	2	XGBoost (ADMETboost features)	0.567	2
tdc_ames	ROC_- AUC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.883	0.912	0.834	7	2	Extra trees	0.871	2
tdc_bbb_martins	ROC_- AUC	Random forest	0.925	0.941	0.903	7	2	XGBoost (ADMETboost features)	0.916	2
tdc_cyp2d6_veith	AUPRC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.763	0.811	0.464	6	2	AdaBoost	0.615	6
tdc_cyp3a4_veith	AUPRC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.905	0.923	0.750	6	2	AdaBoost	0.798	6
tdc_half_life_obach	SPEARMA	AutoMol V1	0.597	0.649	0.485	16	2	XGBoost (ADMETboost features)	0.473	>10

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Dataset	Metric	QSARena best model (test-selected)	QSARena value	Reference top-1	Reference top-10 cutoff	References (n)	Est. rank	CV-selected model	CV-selected value	CV- selected est. rank
tdc_hia_hou	ROC_AUC	XGBoost (ADMETboost features)	0.994	0.994	0.976	8	2	XGBoost (ADMETboost features)	0.994	2
tdc_ld50_zhu	MAE	TabPFNRegressor	0.552	0.292	0.605	16	2	XGBoost	0.588	8
tdc_ppbr_az	MAE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	7.118	0.679	7.914	16	2	TabPFNRegressor	8.537	>10
tdc_solubility_aqsolddb	MAE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.676	0.557	0.776	17	2	XGBoost (ADMETboost features)	0.719	2
tdc_caco2_wang	MAE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.269	0.256	0.288	20	3	XGBoost (ADMETboost features)	0.293	>10
tdc_clearance_hepatocyte_az	SPEARMAN	MANLight + GNN (Cat Boost, Strict Parity)	0.516	0.633	0.440	16	3	XGBoost (ADMETboost features)	0.369	>10
tdc_clintox	ROC_AUC	CFA (Combinatorial Fusion)	0.974	0.996	0.889	12	3	AdaBoost	0.886	>10
tdc_cyp2c9_substrate_carbonmangels	AUPRC	Chemprop v2 (AttentiveFP, ensemble=3)	0.438	0.450	0.360	6	3	SVC	0.301	7
tdc_cyp2c9_veith	AUPRC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.827	0.877	0.770	6	3	AdaBoost	0.705	7
tdc_herg	ROC_AUC	Chemprop v2 (AttentiveFP, ensemble=3)	0.857	0.880	0.806	7	3	XGBoost (ADMETboost features)	0.792	8
tdc_lipophilicity_astrazeneca	MAE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.412	0.406	0.515	17	3	TabPFNRegressor	0.439	4
polaris_adme_fang_hppb_1	MSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.178	0.143	0.303	10	4	XGBoost (ADMETboost features)	0.214	5
polaris_adme_fang_perm_1	MSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.144	0.113	0.239	10	4	XGBoost (ADMETboost features)	0.194	8
tdc_cyp2d6_substrate_carbonmangels	AUPRC	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	0.693	0.766	0.570	7	4	ChemML MLP (PyTorch)	0.512	8
tdc_dili	ROC_AUC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.933	0.945	0.852	6	4	XGBoost (ADMETboost features)	0.923	4
tdc_pgp_broccatelli	ROC_AUC	Uni-Mol V2 (84m)	0.933	0.994	0.911	8	4	Random forest	0.904	9
polaris_adme_fang_rcint_1	MSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.234	0.204	0.337	10	5	XGBoost	0.293	6

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Dataset	Metric	QSARena best model (test-selected)	QSARena value	Reference top-1	Reference top-10 cutoff	References (n)	Est. rank	CV-selected model	CV-selected value	CV- selected est. rank
tdc_vdss_ lombardo	SPEARMAN	MAPLight + GNN (Cat Boost, Strict Parity)	0.680	0.942	0.582	16	5	XGBoost (ADMETboost features)	0.465	>10
tdc_cyp3a4_ substrate_ carbonmangels	ROC_ AUC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.665	0.692	0.651	7	6	XGBoost (ADMETboost features)	0.660	7
esol_delaney	RMSE	Ensemble (Weighted average (inverse OOF error))	0.622	0.558	0.743	30	7	TabPFNRegressor	0.673	9
polaris_adme_ fang_solu_1	MSE	Uni-Mol V2 (84m)	0.304	0.222	0.323	10	7	XGBoost (ADMETboost features)	0.353	>10
tdc_ hydrationfreeenergy_ freesolv	RMSE	Ensemble (Weighted average (inverse OOF error))	1.075	0.654	1.211	12	7	XGBoost (ADMETboost features)	1.138	9
tdc_skin_reaction	ROC_ AUC	XGBoost (ADMETboost features)	0.667	0.741	0.677	21	>10	XGBoost (ADMETboost features)	0.667	>10
tdc_tox21	ROC_ AUC	Chemprop v2 (AttentiveFP, ensemble=3)	0.597	0.867	0.840	12	>10	Extra trees	0.549	>10

Table S8: Applicability domain and calibration on the 22 official TDC ADMET Benchmark Group splits, for a fixed reference model (random forest, 300 trees, on Morgan radius-2 2048-bit fingerprints plus RDKit 2D descriptors) fitted to 80% of each train_val partition, with the remaining 20% used only for conformal calibration. Coverage is the fraction of test compounds inside the domain. The error ratio is the mean test error outside the domain divided by that inside it (MAE for regression, 0/1 error for classification); values above 1 mean the domain flag separates worse predictions. “Std.” is the Roy standardization approach on RDKit descriptors; “kNN” is the 5-nearest-neighbour Tanimoto distance thresholded at the 95th percentile of training distances; “Reliability” is the conformal-interval-width flag (regression) or the posterior-probability confidence flag with $|p_1 - p_0| \geq 0.5$ (classification). Conformal coverage is the empirical test coverage of 90% split-conformal intervals or prediction sets. “—” marks a ratio not reported because fewer than five test compounds fall on one side of the domain boundary. Results describe the reference model, not the benchmark’s selected model; single seed.

Dataset	Task	n test	Std. coverage	Std. ratio	kNN coverage	kNN ratio	Reliability ratio	Conformal coverage	ECE
caco2_wang	reg.	182	0.99	—	0.74	0.75	1.38	0.95	—
clearance_hepatocyte_az	reg.	243	0.94	1.09	0.88	1.26	1.75	0.94	—
clearance_microsome_az	reg.	221	0.99	—	0.97	1.32	2.08	0.93	—
half_life_obach	reg.	135	0.95	1.36	0.94	0.87	3.60	0.94	—
ld50_zhu	reg.	1478	0.95	1.16	0.82	1.24	2.11	0.82	—
lipophilicity_astazeneca	reg.	840	0.99	2.33	0.94	1.55	1.47	0.88	—
ppbr_az	reg.	559	0.99	1.35	0.91	1.05	3.01	0.93	—
solubility_aqsolddb	reg.	1995	0.95	1.45	0.86	1.15	1.59	0.89	—
vdss_lombardo	reg.	226	0.94	0.49	0.86	1.22	8.26	0.95	—
ames	cls.	1457	0.96	0.91	0.84	1.71	2.67	0.89	0.07
bbb_martins	cls.	406	0.96	0.00	0.78	1.19	9.30	0.91	0.06
bioavailability_ma	cls.	128	0.98	—	0.96	0.72	3.72	0.91	0.08
cyp2c9_substrate_carbonmangels	cls.	135	0.93	0.68	0.95	0.00	2.15	0.81	0.11
cyp2c9_veith	cls.	2419	0.96	0.53	0.88	0.62	5.21	0.91	0.05
cyp2d6_substrate_carbonmangels	cls.	135	0.96	0.65	0.98	—	4.82	0.89	0.06
cyp2d6_veith	cls.	2626	0.96	0.90	0.84	0.89	5.29	0.91	0.06
cyp3a4_substrate_carbonmangels	cls.	135	0.95	2.08	0.94	1.80	1.47	0.93	0.06
cyp3a4_veith	cls.	2467	0.98	0.91	0.94	0.93	4.88	0.89	0.07
dili	cls.	96	0.95	3.64	0.95	0.00	2.16	0.95	0.17
herg	cls.	132	0.99	—	0.92	1.10	4.71	0.86	0.10
hia_hou	cls.	117	0.97	—	0.82	4.57	6.13	0.91	0.09
pgp_broccatelli	cls.	245	0.94	0.75	0.84	1.21	5.04	0.88	0.06

Table S9: Dataset meta-feature catalog computed on each benchmark train/test partition (verified against the recorded split hashes). Diversity and similarity use Morgan radius-2, 2,048-bit fingerprints; SNN is the maximum Tanimoto similarity of a test molecule to any training molecule, and OOD fraction is the share of test molecules with SNN below 0.40.

Dataset	Task	Split	Metric	n train	n test	Positive fraction	Imbalance ratio	Target SD	Target skew	Scaffolds	Scaffolds / molecule	Singleton scaffold fraction	Internal diversity	Mean SNN	OOD fraction (SNN < 0.40)
tdc_ames	classification	patidefinedest_-	auc	5,821	1,457	0.533	1.142			2,387	0.410	0.809	0.916	0.517	0.251
		roc_-													
tdc_bbb_martins	classification	patidefinedest_-	auc	1,624	406	0.753	3.050			829	0.510	0.744	0.890	0.418	0.520
		roc_-													
tdc_bioavailability_ma	classification	patidefinedest_-	auc	512	128	0.771	3.376			363	0.709	0.837	0.899	0.435	0.492
		roc_-													
tdc_carcinogens_lagunin	classification	scaffolddest_-	auc	223	57	0.184	4.439			191	0.857	0.864	0.914	0.346	0.684
		roc_-													
tdc_clintox	classification	scaffolddest_-	auc	1,180	298	0.083	11.041			894	0.758	0.813	0.915	0.425	0.517
		roc_-													
tdc_cyp1a2_veith	classification	scaffolddest_-	auc	10,061	2,518	0.445	1.249			7,622	0.758	0.811	0.887	0.561	0.199
		roc_-													
tdc_cyp2c19_veith	classification	scaffolddest_-	auc	10,131	2,534	0.496	1.018			7,615	0.752	0.807	0.887	0.563	0.204
		roc_-													
		auc													

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Dataset	Task	Split	Metric	n train	n test	Positive fraction	Imbalance ratio	Target SD	Target skew	Scaffolds	Scaffolds / molecule	Singleton scaffold fraction	Internal diversity	Mean SNN	OOD fraction (SNN < 0.40)
tdc_cyp2c9_-	classification	patidefined_-	auprc	534	135	0.193	4.184			366	0.685	0.831	0.899	0.408	0.504
substrate_-															
carbonmangels															
tdc_cyp2c9_-	classification	patidefined_-	auprc	9,673	2,419	0.339	1.954			5,727	0.592	0.784	0.885	0.509	0.338
veith															
tdc_cyp2d6_-	classification	patidefined_-	auprc	532	135	0.278	2.595			373	0.701	0.845	0.898	0.426	0.526
substrate_-															
carbonmangels															
tdc_cyp2d6_-	classification	patidefined_-	auprc	10,504	2,626	0.197	4.072			6,643	0.632	0.779	0.884	0.487	0.364
veith															
tdc_cyp3a4_-	classification	patidefined_-	auprc	535	135	0.516	1.066			375	0.701	0.832	0.899	0.436	0.474
substrate_-															
carbonmangels															
tdc_cyp3a4_-	classification	patidefined_-	auprc	9,861	2,467	0.408	1.448			6,088	0.617	0.800	0.889	0.534	0.260
veith															
tdc_dili	classification	patidefined_-	roc_-	379	96	0.491	1.038			265	0.699	0.879	0.907	0.332	0.781
tdc_herg	classification	patidefined_-	auc	523	132	0.677	2.095			341	0.652	0.792	0.895	0.484	0.379
tdc_herg_karim	classification	patidefined_-	auc	10,755	2,690	0.492	1.031			5,884	0.547	0.646	0.878	0.672	0.084

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Dataset	Task	Split	Metric	n train	n test	Positive fraction	Imbalance ratio	Target SD	Target skew	Scaffolds	Scaffolds / molecule	Singleton scaffold fraction	Internal diversity	Mean SNN	OOD fraction (SNN < 0.40)
tdc_hia_hou	classification	patidefinedest_-	auc roc_-	461	117	0.889	8.039			317	0.688	0.868	0.894	0.371	0.598
tdc_pampa_ncats	classification	scaffolddest_-	auc roc_-	1,626	408	0.846	5.504			1,355	0.833	0.862	0.871	0.642	0.064
tdc_pgp_broccatelli	classification	patidefinedest_-	auc roc_-	973	245	0.541	1.177			546	0.561	0.764	0.874	0.490	0.400
tdc_skin_reaction	classification	scaffolddest_-	auc roc_-	289	115	0.661	1.949			247	0.855	0.895	0.918	0.312	0.904
tdc_tox21	classification	scaffolddest_-	auc roc_-	5,797	1,461	0.049	19.412			3,853	0.665	0.872	0.925	0.467	0.305
tdc_toxcast	classification	scaffolddest_-	auc roc_-	1,357	374	0.262	2.812			1,002	0.738	0.889	0.926	0.439	0.422
chemml_cep_homo	regression	target_test_-	auc rmse	400	100			0.687	0.029	400	1	1	0.844	0.492	0.220
chemml_organic_density	regression	target_test_-	auc rmse	400	100			0.031	-0.093	399	0.998	0.997	0.841	0.463	0.250
esol_delaney	regression	scaffolddest_-	auc rmse	874	254			2.247	-0.435	583	0.667	0.873	0.930	0.437	0.417
freesolv_sampler	regression	randomtest_-	rmse	513	129			3.677	-0.753	308	0.600	0.932	0.923	0.570	0.163

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Dataset	Task	Split	Metric	n train	n test	Positive fraction	Imbalance ratio	Target SD	Target skew	Scaffolds	Scaffolds / molecule	Singleton scaffold fraction	Internal diversity	Mean SNN	OOD fraction (SNN < 0.40)
lipophilicity	regression	scaffold	test_rmse	3,357	843			1.190	-0.557	2,355	0.702	0.791	0.882	0.584	0.152
poduam_nc_std	regression	scaffold	test_rmse	1,473	369			1.109	-1.529	945	0.642	0.899	0.922	0.521	0.322
poduam_nc_std	regression	quartiles	test_rmse	1,884	471			0.865	-1.743	1,237	0.657	0.905	0.917	0.549	0.268
polaris_adme_fang_hppb_1	regression	undefined	test_rmse	1,446	362			0.719	-0.337	1,003	0.694	0.827	0.878	0.583	0.331
polaris_adme_fang_perm_1	regression	undefined	test_rmse	2,113	529			0.682	0.746	1,885	0.892	0.926	0.875	0.443	0.465
polaris_adme_fang_rclint_1	regression	undefined	test_rmse	2,443	611			0.751	-0.134	2,146	0.878	0.922	0.876	0.444	0.435
polaris_adme_fang_rppb_1	regression	undefined	test_rmse	708	177			0.751	-0.435	569	0.804	0.872	0.877	0.511	0.429
polaris_adme_fang_solu_1	regression	undefined	test_rmse	1,738	435			0.668	-1.657	1,531	0.881	0.925	0.874	0.446	0.492
tdc_caco2_wang	regression	undefined	test_rmse	728	182			0.797	-0.715	402	0.552	0.694	0.890	0.459	0.473
tdc_clearance	regression	undefined	test_rmse	970	243			50.287	1.216	601	0.620	0.687	0.877	0.425	0.580
hepatocyte_az	regression	undefined	test_rmse	881	221			45.128	1.548	587	0.666	0.818	0.871	0.520	0.321
microsome_az															

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Dataset	Task	Split	Metric	n train	n test	Positive fraction	Imbalance ratio	Target SD	Target skew	Scaffolds	Scaffolds / molecule	Singleton scaffold fraction	Internal diversity	Mean SNN	OOD fraction (SNN < 0.40)
tdc_half_life_obach	regression	undefined	rmse	532	135			90.920	9.338	355	0.667	0.820	0.899	0.413	0.533
tdc_hydrationfreeenergy_freesolv	regression	scaffold	rmse	490	152			4.014	-1.427	381	0.778	0.934	0.927	0.384	0.625
tdc_ld50_zhu	regression	undefined	rmse	5,907	1,478			0.921	0.928	3,136	0.531	0.894	0.917	0.448	0.438
tdc_lipophilicity	regression	undefined	rmse	3,360	840			1.204	-0.582	1,932	0.575	0.769	0.879	0.575	0.219
astrazeneca_tdc_ppbr_az	regression	undefined	rmse	2,231	559			18.785	-1.962	946	0.424	0.498	0.879	0.520	0.349
tdc_solubility_aqsolddb	regression	undefined	rmse	7,985	1,995			2.369	-0.587	4,265	0.534	0.900	0.922	0.475	0.327
tdc_vdss_lombardo	regression	undefined	rmse	904	226			24.829	24.663	613	0.678	0.822	0.900	0.384	0.619

Table S10: Exploratory family $\times$ meta-feature associations: Spearman correlation between each meta-feature and the relative gap of the family’s best model to the per-dataset best, with a dataset-bootstrap 95% CI (10,000 resamples), a two-sided label-permutation p-value and a Benjamini–Hochberg q-value over the whole grid. Positive ρ means the family falls further behind as the meta-feature increases.

Model family	Meta-feature	Datasets	Spearman rho	95% CI	Permutation p	BH q
Conventional ML	log10_n_train	44	0.034	-0.29 to 0.35	0.818	0.873
Conventional ML	internal_diversity	44	-0.329	-0.59 to -0.01	0.028	0.238
Conventional ML	mean_snn	44	0.325	0.01 to 0.58	0.034	0.238
Conventional ML	ood_fraction	44	-0.326	-0.59 to -0.01	0.035	0.238
Conventional ML	scaffolds_per_molecule	44	0.139	-0.20 to 0.43	0.367	0.710
Conventional ML	label_asymmetry	44	0.204	-0.10 to 0.46	0.179	0.461
Ensemble (stacking / averaging)	log10_n_train	44	-0.467	-0.70 to -0.19	0.001	0.058
Ensemble (stacking / averaging)	internal_diversity	44	0.207	-0.14 to 0.52	0.180	0.461
Ensemble (stacking / averaging)	mean_snn	44	-0.393	-0.66 to -0.09	0.008	0.202
Ensemble (stacking / averaging)	ood_fraction	44	0.288	-0.03 to 0.59	0.062	0.297
Ensemble (stacking / averaging)	scaffolds_per_molecule	44	0.147	-0.14 to 0.43	0.336	0.710
Ensemble (stacking / averaging)	label_asymmetry	44	0.275	-0.03 to 0.54	0.070	0.308
CFA combinatorial fusion	log10_n_train	44	-0.220	-0.47 to 0.07	0.156	0.461
CFA combinatorial fusion	internal_diversity	44	-0.325	-0.59 to -0.01	0.033	0.238
CFA combinatorial fusion	mean_snn	44	0.125	-0.21 to 0.43	0.421	0.710
CFA combinatorial fusion	ood_fraction	44	-0.081	-0.39 to 0.24	0.602	0.743
CFA combinatorial fusion	scaffolds_per_molecule	44	0.216	-0.10 to 0.50	0.159	0.461
CFA combinatorial fusion	label_asymmetry	44	0.108	-0.19 to 0.40	0.486	0.710
Uni-Mol (3D pretrained)	log10_n_train	44	-0.108	-0.41 to 0.20	0.488	0.710
Uni-Mol (3D pretrained)	internal_diversity	44	0.146	-0.21 to 0.49	0.348	0.710
Uni-Mol (3D pretrained)	mean_snn	44	0.081	-0.25 to 0.40	0.604	0.743
Uni-Mol (3D pretrained)	ood_fraction	44	-0.118	-0.46 to 0.22	0.442	0.710
Uni-Mol (3D pretrained)	scaffolds_per_molecule	44	0.102	-0.23 to 0.41	0.516	0.729
Uni-Mol (3D pretrained)	label_asymmetry	44	0.075	-0.26 to 0.41	0.628	0.753
TabPFN (tabular foundation)	log10_n_train	41	0.149	-0.24 to 0.50	0.346	0.710
TabPFN (tabular foundation)	internal_diversity	41	-0.242	-0.54 to 0.10	0.130	0.461
TabPFN (tabular foundation)	mean_snn	41	0.098	-0.21 to 0.40	0.540	0.730
TabPFN (tabular foundation)	ood_fraction	41	-0.004	-0.34 to 0.33	0.983	0.983
TabPFN (tabular foundation)	scaffolds_per_molecule	41	0.071	-0.28 to 0.41	0.661	0.755
TabPFN (tabular foundation)	label_asymmetry	41	-0.126	-0.42 to 0.18	0.428	0.710
Chemprop v2 GNN	log10_n_train	42	-0.008	-0.31 to 0.30	0.965	0.983
Chemprop v2 GNN	internal_diversity	42	-0.050	-0.37 to 0.28	0.752	0.840
Chemprop v2 GNN	mean_snn	42	0.131	-0.19 to 0.43	0.405	0.710
Chemprop v2 GNN	ood_fraction	42	-0.119	-0.44 to 0.21	0.454	0.710

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Model family	Meta-feature	Datasets	Spearman rho	95% CI	Permutation p	BH q
Chemprop v2 GNN	scaffolds_per_molecule	42	0.141	-0.16 to 0.42	0.370	0.710
Chemprop v2 GNN	label_asymmetry	42	0.325	-0.03 to 0.62	0.031	0.238
MapLight + GNN	log10_n_train	44	-0.205	-0.47 to 0.10	0.182	0.461
MapLight + GNN	internal_diversity	44	0.091	-0.23 to 0.40	0.559	0.730
MapLight + GNN	mean_snn	44	-0.071	-0.38 to 0.24	0.643	0.753
MapLight + GNN	ood_fraction	44	-0.113	-0.42 to 0.22	0.467	0.710
MapLight + GNN	scaffolds_per_molecule	44	0.291	0.01 to 0.54	0.052	0.278
MapLight + GNN	label_asymmetry	44	-0.090	-0.39 to 0.25	0.563	0.730
Deep tabular NN (ChemML MLP)	log10_n_train	44	-0.315	-0.57 to 0.00	0.042	0.250
Deep tabular NN (ChemML MLP)	internal_diversity	44	-0.251	-0.51 to 0.04	0.100	0.401
Deep tabular NN (ChemML MLP)	mean_snn	44	-0.017	-0.33 to 0.31	0.912	0.952
Deep tabular NN (ChemML MLP)	ood_fraction	44	0.039	-0.31 to 0.36	0.807	0.873
Deep tabular NN (ChemML MLP)	scaffolds_per_molecule	44	0.224	-0.10 to 0.51	0.141	0.461
Deep tabular NN (ChemML MLP)	label_asymmetry	44	0.181	-0.14 to 0.47	0.242	0.581

Table S11: Family selector (v2), leave-one-dataset-out regret: the relative gap (%) of the recommended family to the per-dataset best. Rows are pre-specified variants (model | feature blocks: F0 size, similarity, label asymmetry and task; F1 chemistry; F2 label landscape; F3 training-set CV landmarks); “sbs” always picks the family with the best record on the other datasets. The nested row chooses the variant inside an inner leave-one-dataset-out loop and is the only unbiased estimate; single variants are exploratory.

Selector (model feature blocks)	Mean regret (%)	Median regret (%)	Pick within 5% of best (%)	SBS-to-oracle gap closed
sbs	3.310	0.031	84.091	0
knn F0	3.760	0.159	79.545	-0.136
ridge F0	2.982	0	86.364	0.099
rf F0	4.236	0.082	77.273	-0.280
knn F0+F1	3.629	0	81.818	-0.096
ridge F0+F1	3.403	0	81.818	-0.028
rf F0+F1	3.650	1.086	81.818	-0.102
knn F0+F2	3.728	0.159	81.818	-0.126
ridge F0+F2	3.323	0	84.091	-0.004
rf F0+F2	4.163	0.159	75	-0.257
knn F0+F3	2.739	0	88.636	0.173
ridge F0+F3	3.133	0	84.091	0.054
rf F0+F3	3.216	0	84.091	0.028
knn F0-F3	3.375	0	81.818	-0.020
ridge F0-F3	4.525	0.031	81.818	-0.367
rf F0-F3	3.516	0.051	81.818	-0.062
tree_cls F0	4.837	0	79.545	-0.461
nested (headline)	3.115	0	86.364	0.059

Supplementary Note S1: Benchmark runs, ensemble reconstruction and computing environment

This note gives the full provenance that main-text Section 2.12 summarizes, including the run directories, commit and commands, which the main text leaves out.

The benchmark reported here was executed on NSF ACCESS Jetstream2 [1, 2] under allocation CIS261142, on a `g3.large` instance providing one NVIDIA A100-SXM4-40GB GPU (39.5 GB device memory, 108 streaming multiprocessors, compute capability 8.0) and 32 CPU cores, running Linux with all timestamps in UTC. The device inventory, CPU count, parallelism settings and `fp32` precision mode are recorded in the run’s `run_timing.json` and `run_config.json` and are reproduced in the deposited artifacts. The Chemprop and TabPFN repair run used the same node; the out-of-fold ensemble rebuild ran on the RTX 4060 workstation (main-text Section 2.12), and the wall-clock and cost figures in this paper come from the original A100 run. Main-text Section 3.10 additionally reports the same benchmark executed on a consumer laptop GPU, as a direct test of whether the results depend on datacentre hardware.

All figures, tables and numerical claims in this paper are regenerated from committed benchmark artifacts by a single command, `python portable_colab_qsar_bundle/render_manuscript_assets.py`, which re-executes the analysis notebook and writes the figures, tables and a `manuscript_numbers.json` containing every headline number quoted in the text. A companion script, `verify_manuscript_numbers.py`, asserts that the manuscript text still matches those artifacts.

Every reported result is backed by a complete artifact trail. Each analysed dataset retains metrics, selected-feature records, selector coefficients, split-signature hashes, per-stage runtimes, CFA candidate tables and ensemble weights; the run recorded a configuration signature, fixed seeds (random seed 13, Chemprop seed 42), a full benchmark profile, `fp32` precision and a GPU inventory identifying the A100 device; and a SHA-256 manifest covers repository-level and per-dataset artifacts. The benchmark artifacts analysed here were committed in repository revision `bbfb188`. All figures, tables and quoted numbers in this paper regenerate from those artifacts with one command (main-text Section 2.11), which also writes a machine-readable record of every headline number, and a companion script asserts that the manuscript text still matches it.

The reported results combine three runs that share one data, split, feature and feature-selection configuration, which the runner verifies through stage signatures. (i) The base benchmark ran once on the Jetstream2 A100 node described in main-text Section 2.11 (`benchmark_results/autoqsar_benchmark_20260623_153839`). In that run, a text-encoding fault in the Chemprop harness discarded most Chemprop results, and TabPFN was disabled. (ii) A repair run on the same node (`benchmark_results/qsarena_benchmark_chemprop_fixed`) was seeded with the first run’s per-model results and trained only what was missing: the five Chemprop variants, which then produced valid results on 38 to 42 of the 44 datasets, TabPFN (valid on 31 of 44 before the API’s daily limit stopped it) and one interrupted dataset. No base model was retrained. (iii) Every ensemble was then rebuilt from out-of-fold predictions (`benchmark_results/qsarena_benchmark_oof_ensemble`, from which all reported results are computed). Each candidate member received an out-of-fold prediction for every training molecule on the training split’s five cross-validation folds. Uni-Mol supplied the predictions saved from its own internal five-fold training; every other member was refitted on each fold with its reported configuration, on a workstation with a consumer NVIDIA RTX 4060 laptop GPU. Base-model results are unchanged from the source runs, which the release checks automatically, with the documented exceptions below. (iv) The descriptor model of main-text Section 2.6 was trained into this run with the same configuration as every other model, after it performed well in an exploratory analysis. We flag this because adding a model after seeing results can flatter test-selected comparisons. (v) Four binary datasets are catalogued with a regression metric (CYP1A2, CYP2C19, hERG-Karim and PAMPA). On them, ten conventional and gradient-boosting models saved hard class labels instead of probabilities, so their test AUROC was computed on labels. Those models were retrained on those datasets with probability outputs, which raised their test AUROC by a median of 0.06 to 0.11 per dataset. TabPFN was retrained on PAMPA with the locally installed TabPFN package (version 2.6 weights); on the other three datasets its label-based results are withdrawn, because the local model does not fit the

workstation GPU at their size (about 10,000 training molecules and 1,000 selected features). Rerunning them through the Prior Labs API, which produced the other TabPFN results, was not feasible either: the API is metered by tokens (training rows times features times model fits), and the runner’s estimate for one run of these datasets with cross-validation is 111 to 127 million tokens each, more than twenty times the API’s daily allowance of 5 million tokens per key. TabPFN therefore has no result on these three datasets. (vi) Finally, the cross-validated metrics and out-of-fold predictions of every model that uses the selected features were recomputed with feature selection nested inside each fold (main-text Section 2.5), and every ensemble was rebuilt once over the corrected 31-model pool. TabPFN is not an ensemble member, because most of its full-fit predictions came from the Prior Labs API while its fold refits used the local package. On hERG and bioavailability, the conventional and ChemML models have no saved predictions, so they receive nested cross-validated scores but are not ensemble members there. Combinatorial fusion uses no cross-validated scores and was rebuilt once over the 31-model pool.

The rebuild replaced two earlier ensemble protocols. The first run selected and weighted ensemble members by held-out test metrics, which leaks test information into the ensemble row. A second attempt used in-sample training predictions instead. That removed the leak but rewarded memorisation: tree ensembles that fit the training set almost exactly took nearly all of the stacking weight, and regression ensemble wins fell from 7 to 1. Under the out-of-fold protocol with nested feature selection, ensembles won 22 datasets (14 regression, 8 classification), against 16 (7 and 9) under the leaky first protocol and 12 (6 and 6) when their out-of-fold predictions came from the outer feature selection. The feature-selection leak had also inflated the out-of-fold predictions of the members that use selected features, so the stacker over-weighted them; nesting the selection improved the best ensemble on 39 of the 44 datasets (median error reduction 4.9% for regression; median AUROC gain 0.7% for classification), while no base model’s test result changed. The rebuild also fixed two faults that would otherwise have biased the ensembles. Uni-Mol’s saved fold predictions are on a normalised target scale, and its scaler file was not among the transferred artifacts, so the scaler is now rebuilt from the training targets. And a few members extrapolated far outside the target range on single molecules, which motivated the clipping described in main-text Section 2.7. Two Chemprop variants still lack out-of-fold predictions on Tox21, where Chemprop cannot featurise one training molecule, and are excluded there.

Two datasets need a note on coverage. A 45th dataset, `tdc_herg_central`, was abandoned for compute budget after more than 24 hours in the base run without completing, and is not analysed. `polaris_adme_fang_hppb_1` was interrupted when the base run ended and was completed by the repair run, so all 44 analysed datasets are complete.

Supplementary Note S2: Molecular representations, feature store and model configurations

Ten configurable feature families were implemented on RDKit, all returning fixed-width numeric matrices. Circular and path-based fingerprints were generated at 1024 bits by default. Morgan fingerprints [3] used radius 2; ECFP6 and FCFP6 were generated with the RDKit MorganGenerator at radius 3, FCFP6 using feature-based atom invariants; RDKit layered, atom-pair, topological-torsion and RDKit path fingerprints were each computed at 1024 bits, the path fingerprint spanning path lengths 1–7. MACCS keys [4] were generated at their native 167-bit width, and the full RDKit 2D descriptor set was computed from `Descriptors._descList`.

The “MapLight classic” family reproduces the feature construction of the MapLight TDC submission [5]: a hashed Morgan count fingerprint (radius 2, 1024 bits), an Avalon count fingerprint (1024 bits) [6], the extended reduced graph (ErG) fingerprint [7], and a curated panel of approximately 200 RDKit descriptors spanning connectivity and shape indices (BalabanJ, BertzCT, Chi and Kappa indices), VSA-family descriptors, physicochemical properties (ExactMolWt, MolLogP, TPSA, FractionCSP3), hydrogen-bonding and ring counts, functional-group fragment counts, and the QED drug-likeness score. In the feature-selection analyses below, this composite appears as its three constituent families (`avalon`, `erg` and the `maplight` descriptor panel), and we report both the components and their sum.

After assembly, every feature matrix passed a common finalization step: infinities replaced with missing values, all columns coerced to numeric, values exceeding the float32 range masked, remaining gaps imputed with the per-column median (falling back to zero), and the matrix cast to float32.

To avoid recomputing expensive descriptors across datasets and runs, features were cached in a persistent, content-addressed store writing Parquet shards where `pyarrow` or `fastparquet` is available and CSV otherwise. Each representation is identified by a key derived from a SHA-256 hash of a canonical JSON payload listing the selected feature families, Morgan radius and fingerprint bit width, so distinct configurations never collide. A schema file records the representation payload and column order; a mismatch raises an error rather than silently mixing definitions. Feature rows are keyed by canonical SMILES, so only molecules absent from the cache are computed on a given run, and cached and new rows are merged back into the exact requested row order. Additional caches store assembled benchmark feature matrices and tuned conventional models.

For TDC datasets, official splits were preferred over generic resampling. Where PyTDC exposes an official `admet_group` entry, the runner uses that `train_val/test` split in preference to legacy single-prediction cache entries; official split frames are cached under `data/_qsarena_cache` and stale entries lacking an official split are refreshed automatically. This distinction matters for interpretation and we preserve it throughout: 22 TDC datasets and 5 Polaris datasets used official predefined splits, while the remaining 18 used locally generated scaffold, random or target-quartile splits and are therefore *not* directly leaderboard-equivalent (main-text Section 3.5).

Selection was then performed train-only with a cross-validated elastic net. The ElasticNetCV selector was fit in an isolated subprocess with a wall-clock timeout over a configurable L1-ratio grid and log-spaced alpha grid, using folds matched to the active split strategy. Features were retained when the absolute coefficient exceeded a threshold, with at least one feature always kept and the retained set capped at the larger of one feature or 10% of the training-row count. To bound runtime, the selector used a dataset-size runtime model: when predicted elastic-net runtime exceeded a configurable threshold (default 7200s), or when the elastic net timed out or failed, selection fell back to RandomForest importance ranking. Selected features, coefficients and diagnostics were written per dataset.

Conventional machine learning. For regression: elastic-net regression with internal cross-validated alpha/L1 search, support vector regression ($C = 10$, $\epsilon = 0.1$, RBF kernel), random forests (400 trees), extremely randomized trees (500 trees), histogram gradient boosting (learning rate 0.05, up to 500 iterations, maximum depth 8), a soft-voting KNN+SVR regressor with adaptively chosen neighbour count, AdaBoost (500 estimators, learning rate 0.05), and a tabular multilayer perceptron (hidden layers 512 and 256, ReLU, Adam, L2 1×10^{-4} , up to 300 iterations), all via scikit-learn [8]. Gradient-boosting libraries were added when installed: XGBoost (400 trees, depth 6,

learning rate 0.05, subsample and column-sample 0.9) [9], LightGBM (500 trees, learning rate 0.05, 63 leaves) [10] and CatBoost (400 iterations, depth 6, learning rate 0.05) [11]. Numeric pipelines were preceded by median imputation and standardization where appropriate. The classification table substituted the analogous estimators (logistic regression, SVC with probability estimates, random forest, extra trees, histogram gradient boosting, soft-voting KNN+SVC, AdaBoost, tabular MLP and the gradient-boosting classifiers) with classification losses and metrics. One further gradient-boosting model, XGBoost on the full, unselected ADMETboost feature set (MACCS keys, ECFP4 and PubChem fingerprints, Mol2Vec embeddings, and Mordred 2D and RDKit 2D descriptors; 1,000 trees, learning rate 0.05, maximum depth 6, row and column subsampling 0.8 and 0.5), bypasses feature selection altogether. It joined the library after the main run (main-text Section 2.12).

Specialized tabular and graph models. A compact one-dimensional convolutional regressor treats each standardized feature vector as a 1D signal and applies two same-padded ReLU convolutional blocks (64 filters, kernel size 5), global max pooling, a 128-unit dense layer with dropout and a linear output, trained with Adam and early stopping; it is intentionally small enough to run on CPU. The MapLight + GNN workflow fits a CatBoost model on the union of MapLight classic features and pretrained graph isomorphism network (GIN) fingerprints where the supporting DGL/PyTorch stack is available; a strict leaderboard-parity variant of the MapLight CatBoost model uses mean-absolute-error optimization, target scaling and five-seed averaging to reproduce the published protocol [5]. Graph neural networks were provided through Chemprop v2 [12, 13, 14] with configured directed message-passing (D-MPNN), CMPNN-style and AttentiveFP-style architectures; optional variants augment the graph encoder with train-only selected tabular descriptors or an RDKit-2D featurizer. Uni-Mol V1 and V2 [15, 16] were included as 3D pretrained baselines, running automatically when a GPU was detected. The configuration that produced the deposited results was 20 epochs, learning rate 1×10^{-4} , batch size 32, early stopping patience 5, a random internal validation split, a 64-atom cap and the 84m checkpoint, with automatic mixed precision enabled under an otherwise fp32 precision mode. These values are recorded in `run_config.json` and in the per-model `unimol_*` columns of every `metrics.csv`. The TabPFN tabular foundation model [17] was available for regression and classification; most of its results came from the Prior Labs API and some refits from the locally installed package (Note S1).

After base models produced aligned train and test prediction vectors, QSARena optionally fused them by combinatorial fusion analysis (CFA) [18] in both score and rank spaces. To control combinatorial growth, inputs were first reduced to the best model per workflow and the subset search bounded by a budget guardrail. For each candidate subset the algorithm computed a performance strength as the inverse of the base model’s training error and a diversity strength from the mean pairwise distance between normalized, sorted score profiles, then evaluated three score-space weightings (equal, performance-weighted, diversity-weighted) plus the corresponding rank-space variants. Rank-space combinations were linearly calibrated back to the target scale and given a small metric discount when subset diversity exceeded a threshold, so that diverse rank fusions were preferred only when genuinely complementary. Candidates were ranked by the adjusted training metric and the best fused predictor returned with its selected models, weights and a candidate-diagnostics table. For regression, fusion minimized mean absolute error.

Table S12 lists every model in the library with its per-dataset availability.

Table S12: Model inventory. “Valid” datasets are those on which the model produced a metric without an error; differences from 44 reflect task-type applicability (regression-only or classification-only estimators), dataset-size guardrails, or backend failures (main-text Section 3.9).

Model family	Model	Valid regression datasets	Valid classification datasets	Wins	Datasets attempted
CFA combinatorial fusion	CFA (Combinatorial Fusion)	22	22	1	44
Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=3)	20	18	4	44
Chemprop v2 GNN	Chemprop v2 (CMPNN, ensemble=3)	22	20	0	44

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Model family	Model	Valid regression datasets	Valid classification datasets	Wins	Datasets attempted
Chemprop v2 GNN	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	20	18	1	44
Chemprop v2 GNN	Chemprop v2 (D-MPNN + Selected descriptors, ensemble=3)	22	20	0	44
Chemprop v2 GNN	Chemprop v2 (D-MPNN, ensemble=3)	22	20	0	44
Conventional ML	AdaBoost	22	22	0	44
Conventional ML	CatBoost	22	22	1	44
Conventional ML	ElasticNetCV	20	0	0	22
Conventional ML	Extra trees	22	22	0	44
Conventional ML	HistGradientBoosting	22	22	1	44
Conventional ML	LogisticRegression	0	22	0	22
Conventional ML	MapLight CatBoost (Strict Parity)	22	0	0	22
Conventional ML	Random forest	22	22	1	44
Conventional ML	SVC	0	22	0	22
Conventional ML	SVR	22	0	0	22
Conventional ML	Tabular CNN	22	0	0	22
Conventional ML	Tabular MLP	22	22	0	44
Conventional ML	Voting Classifier (KNN, SVM)	0	22	0	22
Conventional ML	Voting Regressor (KNN, SVM)	22	0	0	22
Conventional ML	XGBoost	22	22	0	44
Conventional ML	XGBoost (ADMETboost features)	22	22	3	44
Deep tabular NN (ChemML MLP)	ChemML MLP (PyTorch)	22	22	0	44
Deep tabular NN (ChemML MLP)	ChemML MLP (TensorFlow)	22	22	0	44
Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	22	22	18	44
Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse OOF error))	22	22	4	44
MapLight + GNN	MapLight + GNN (CatBoost, Strict Parity)	22	22	2	44
TabPFN (tabular foundation)	TabPFNClassifier	0	19	0	22
TabPFN (tabular foundation)	TabPFNRegressor	22	0	3	22
Uni-Mol (3D pretrained)	Uni-Mol V1	22	22	2	44
Uni-Mol (3D pretrained)	Uni-Mol V2 (84m)	10	7	3	44

Supplementary Note S3: Sensitivity of the family comparison

Both analyses in this note read only the committed per-dataset metric tables of the manuscript run; no model was retrained. They are produced by `reanalysis_coverage_posthoc.py`, which first reproduces every family’s best model per dataset exactly as the main analysis reports it, and their numbers are checked against the manuscript by the verification script.

Common-subset comparison. Model coverage is uneven (Table S4): Chemprop v2 produced valid results on 38 to 42 of the 44 datasets depending on variant, TabPFN on 41, and Uni-Mol V2 on 17 (Uni-Mol V1 on all 44, so the Uni-Mol family has a result everywhere). Table S13 recomputes each family’s wins, median relative gap to the per-dataset best and share of datasets within 5% of the best on all 44 datasets, on the 41 datasets on which all eight families produced a result, and on the 17 datasets on which Uni-Mol V2 ran. Restricting the dataset set does not change any dataset’s winner, which is chosen over all models on that dataset; it changes which datasets are counted. The single-family win counts are unchanged on the 41-dataset subset, and conventional machine learning keeps the smallest median gap of any single family on both subsets, so the comparison between conventional and pretrained families does not depend on which datasets each family covered.

Post-hoc descriptor model. XGBoost on the full, unselected ADMETboost feature set joined the library after it performed well in an exploratory analysis (main-text Section 2.12). Adding a model after seeing results can flatter test-selected comparisons, so Table S14 recomputes the win tally with that model removed as a candidate. It won three datasets; without it, two of them go to the out-of-fold stacking ensemble and one to Uni-Mol V2, so conventional machine learning falls from 6 wins to 3, and no single family wins more than 6. The ensemble rows still contain the descriptor model as a member on the datasets where it qualified; removing it there would need an ensemble rebuild, which was not done, so the table bounds only its direct effect.

Table S13: Common-subset sensitivity of the family comparison: wins, median relative gap of each family’s best model to the per-dataset best, and share of datasets within 5% of the best, on all datasets, on the datasets on which all eight families produced a valid result, and on the datasets on which Uni-Mol V2 produced one. Winners are chosen over all models on each dataset, so restricting the dataset set changes which datasets are counted, not any dataset’s winner. Single split and single seed.

Dataset subset	Model family	Datasets	Wins	Median gap to best (%)	Within 5% of best (%)
All datasets (n = 44)	Ensemble (stacking / averaging)	44	22	0.0	84.0
All datasets (n = 44)	Conventional ML	44	6	3.2	66.0
All datasets (n = 44)	Chemprop v2 GNN	42	5	5.6	48.0
All datasets (n = 44)	Uni-Mol (3D pretrained)	44	5	5.4	48.0
All datasets (n = 44)	TabPFN (tabular foundation)	41	3	8.4	22.0
All datasets (n = 44)	MapLight + GNN	44	2	12.5	11.0
All datasets (n = 44)	CFA combinatorial fusion	44	1	5.2	50.0
All datasets (n = 44)	Deep tabular NN (ChemML MLP)	44	0	16.6	18.0
All eight families valid (n = 41)	Ensemble (stacking / averaging)	41	19	0.1	83.0
All eight families valid (n = 41)	Conventional ML	41	6	3.7	63.0
All eight families valid (n = 41)	Chemprop v2 GNN	41	5	5.9	46.0
All eight families valid (n = 41)	Uni-Mol (3D pretrained)	41	5	5.8	44.0
All eight families valid (n = 41)	TabPFN (tabular foundation)	41	3	8.4	22.0
All eight families valid (n = 41)	MapLight + GNN	41	2	12.8	10.0
All eight families valid (n = 41)	CFA combinatorial fusion	41	1	5.8	46.0
All eight families valid (n = 41)	Deep tabular NN (ChemML MLP)	41	0	17.7	12.0
Uni-Mol V2 valid (n = 17)	Ensemble (stacking / averaging)	17	8	0.2	76.0

continued on next page

Dataset subset	Model family	Datasets	Wins	Median gap to best (%)	Within 5% of best (%)
Uni-Mol V2 valid (n = 17)	Conventional ML	17	2	3.7	53.0
Uni-Mol V2 valid (n = 17)	Chemprop v2 GNN	17	1	7.6	47.0
Uni-Mol V2 valid (n = 17)	Uni-Mol (3D pretrained)	17	4	5.4	47.0
Uni-Mol V2 valid (n = 17)	TabPFN (tabular foundation)	16	2	9.5	31.0
Uni-Mol V2 valid (n = 17)	MapLight + GNN	17	0	13.9	12.0
Uni-Mol V2 valid (n = 17)	CFA combinatorial fusion	17	0	5.9	35.0
Uni-Mol V2 valid (n = 17)	Deep tabular NN (ChemML MLP)	17	0	20.1	12.0

Table S14: Datasets won by each family with and without the post-hoc descriptor model (XGBoost on the full, unselected ADMETboost feature set) as a candidate. The ensemble rows are unchanged models that still contain it as a member; removing it there would need an ensemble rebuild, so the table bounds only its direct effect.

Model family	Wins with the descriptor model	Wins without it	Change
Ensemble (stacking / averaging)	22	24	2
Conventional ML	6	3	-3
Chemprop v2 GNN	5	5	0
Uni-Mol (3D pretrained)	5	6	1
TabPFN (tabular foundation)	3	3	0
MapLight + GNN	2	2	0
CFA combinatorial fusion	1	1	0
Deep tabular NN (ChemML MLP)	0	0	0

Supplementary Note S4: When does each family win? A dataset-property meta-analysis

Section 3.2 of the main text found that no model family dominates. Here we ask when each family comes close to the best, using properties of the dataset alone. This is per-dataset algorithm selection from dataset meta-features, the approach Meta-QSAR applied to thousands of QSAR problems [19], and it addresses the observation that the best ADMET model and representation are strongly dataset-dependent [20]. For each of the 44 datasets we computed meta-features on the exact train/test partition used in the benchmark, verified against the recorded split hashes (Table S9). They are: training-set size, label imbalance or skew, Bemis–Murcko scaffold diversity, internal fingerprint diversity, and train-to-test similarity. Similarity is summarised by SNN, the Tanimoto similarity of each test molecule to its nearest training molecule, which is known to track prediction error [21]. Winners are single-split, single-seed outcomes, so the outcome modelled is the continuous relative gap of main-text Figure 6, never the identity of the winner. Every interval is a 95% percentile bootstrap over datasets.

Training-set size. The gap of the best conventional-ML model to the per-dataset best showed no clear monotone trend with training-set size (Spearman $\rho = 0.03$, 95% CI -0.29 to 0.35 , BH $q = 0.87$). For the pre-registered contrast between conventional ML and the 3D-pretrained Uni-Mol family (Figure S1), the fitted gap curves do not cross inside the observed size range (a crossing occurred in only 30% of dataset-bootstrap replicates). Tuned comparisons of D-MPNN and random forest report crossovers at roughly 500–2,000 training compounds [22], and dataset size is known to govern when representation learning pays off [23]. Under QSARena’s single fixed configuration, no such crossover is resolved: neither family’s fitted gap overtakes the other within the observed sizes.

Chemical-space shift. Mean SNN was not clearly associated with the achievable best held-out metric (within-task rank; Spearman $\rho = 0.17$, 95% CI -0.14 to 0.48 , permutation $p = 0.262$; Figure S2a). Of the screened properties, the out-of-domain fraction (test molecules with SNN < 0.40) had the strongest association with achievable performance ($\rho = -0.26$, 95% CI -0.57 to 0.08), although its interval also includes zero. Message-passing models have been reported to generalise to unseen chemical space better than tree-based models [12], and the 7 datasets re-split to scaffold splits between our two runs (main-text Section 3.10) offer a small natural experiment on identical chemistry (Figure S2b). Moving them to scaffold splits changed the median family gap by +1.8 percentage points for conventional ML and +1.1 for Uni-Mol. Families valid in both runs on fewer than 3 of them are not summarised: Chemprop (1 of 7). With 7 datasets, and with the two runs also differing in hardware and model settings, this is descriptive only.

Family $\times$ property grid. Across 48 family $\times$ meta-feature correlations with Benjamini–Hochberg control (Table S10), one survived at $q < 0.10$, the strongest being Ensemble (stacking / averaging) versus log10 training-set size ($\rho = -0.47$, 95% CI -0.70 to -0.19 , $q = 0.06$).

A family recommender. A leave-one-dataset-out depth-3 decision tree used four meta-features: log10 training-set size, mean SNN, label asymmetry and task. It predicted the best family group (fusion, descriptor-based ML, Uni-Mol or Chemprop) with balanced accuracy 0.32 (95% CI 0.23 to 0.41). The majority-class baseline scored 0.25, and label permutation scored 0.24 (permutation $p = 0.138$; Figure S3). Its interval overlaps the baselines, so dataset meta-features alone do not reliably identify the winning family at this sample size; the per-dataset winner remains an empirical question, which is the case for benchmarking many families on every dataset.

Predicting the winner’s identity is a harsh test, because many datasets have several families within a few percent of each other. Algorithm selection is usually judged instead by regret, the performance given up by the recommended choice [19]. Here regret is the gap of the recommended family, and the selector was designed and fixed before any result was seen. Each held-out dataset’s selector was chosen by an inner leave-one-dataset-out loop among 12 variants. These were nearest-dataset and per-family ridge models over size, similarity, chemistry, label-landscape and training-set cross-validation landmark features. Nested selection did not reliably beat the single best family (mean regret 3.1% versus 3.3% for always choosing the family with the best average record (difference 0.2 percentage points, 95% CI -0.9 to 1.3 ; permutation $p = 0.099$)). Both lose little: a randomly chosen valid family would cost 10.7% on average. The picked family was within 5% of the best on 86% of datasets, against 84% for

the single best family. The inner loop most often chose the nearest-datasets model on F0+F3 (66% of held-out datasets; all variants in Table S11).

Diversity metrics depend on the fingerprint. Recomputed with ECFP6 at 4,096 bits, the fingerprint-dependent correlations kept their sign in 92% of cells and their significance classification in 100%. These relationships are exploratory. They rest on one split and one seed per dataset and on $n = 44$ datasets, so they generate hypotheses for a multi-seed or learning-curve study; they do not establish selection rules.

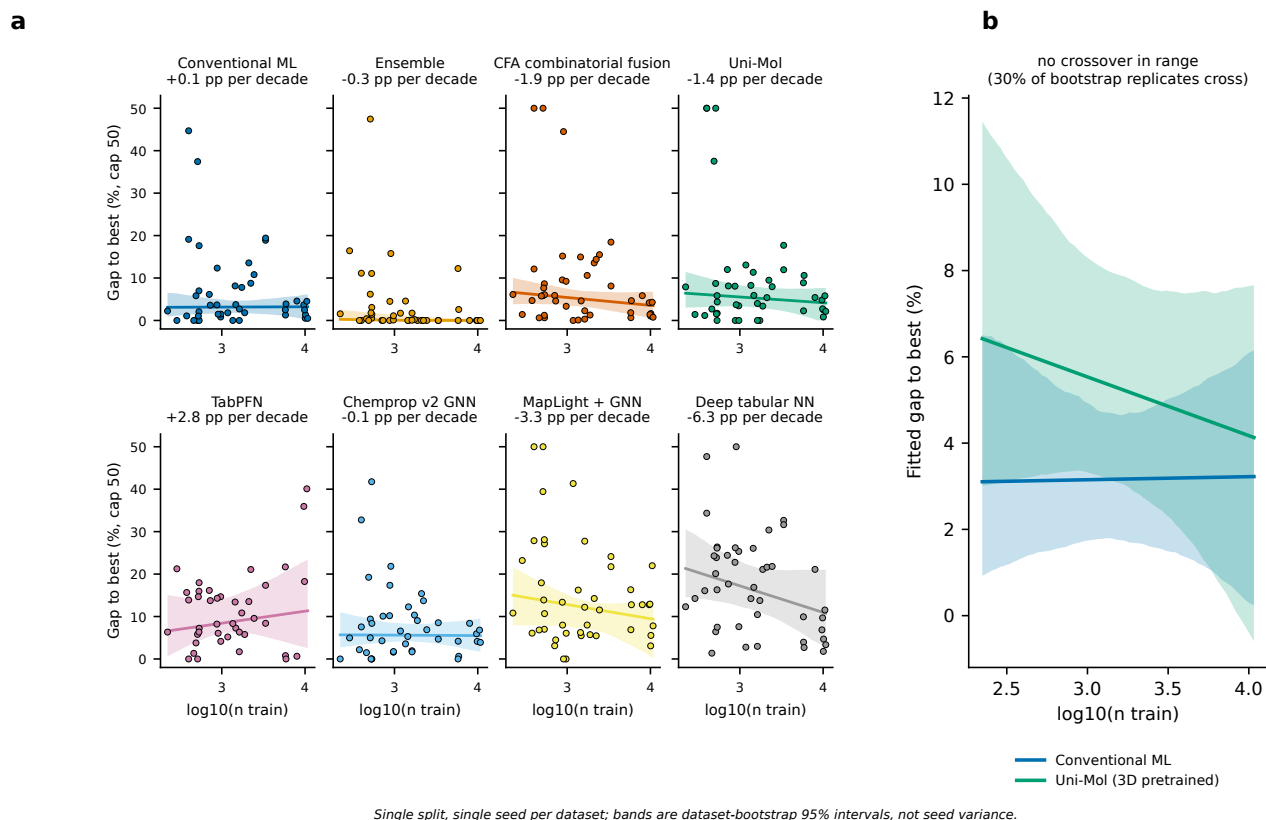


Figure S1: (a) Relative gap of each family's best model to the per-dataset best versus log10 training-set size, with Theil-Sen fits and dataset-bootstrap 95% bands (gaps capped at 50% for display). (b) Fitted gap curves for conventional ML and Uni-Mol; the dashed line and grey span mark the estimated crossover and its bootstrap CI when one exists. Single split and single seed per dataset: bands reflect dataset resampling, not seed variance.

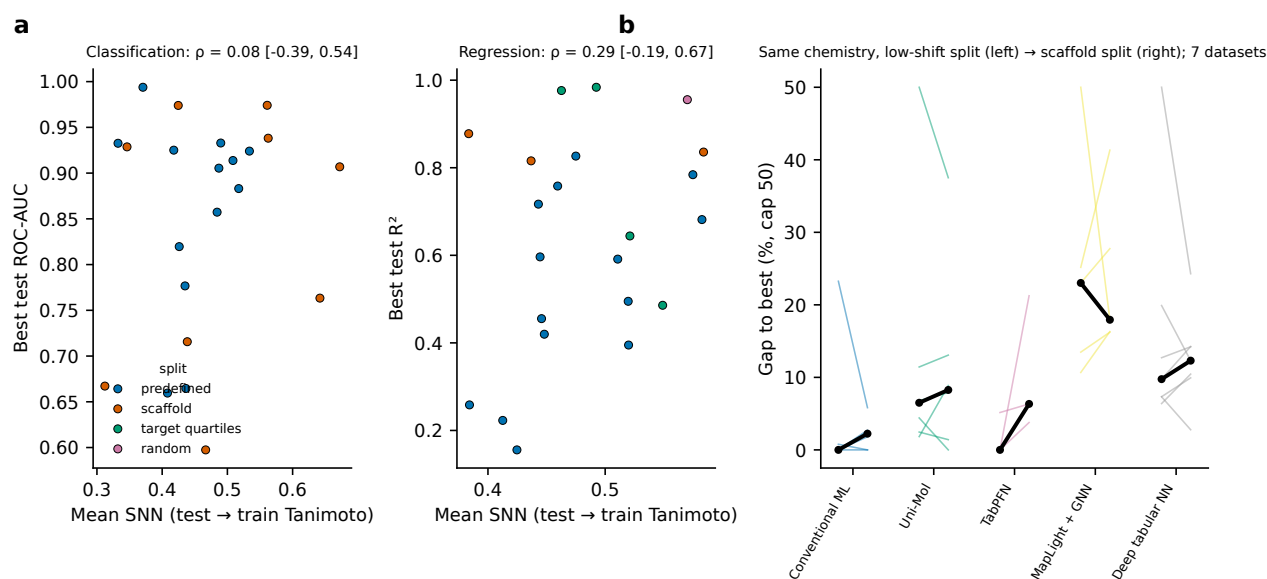


Figure S2: (a) Achievable best held-out metric versus mean test-to-train nearest-neighbour Tanimoto (SNN), by task and split protocol. (b) Family gaps on the 7 datasets that were re-split from random or target-quartile splits to scaffold splits between the RTX 4060 and A100 runs (main-text Section 3.10); thin lines are datasets, thick lines are medians.

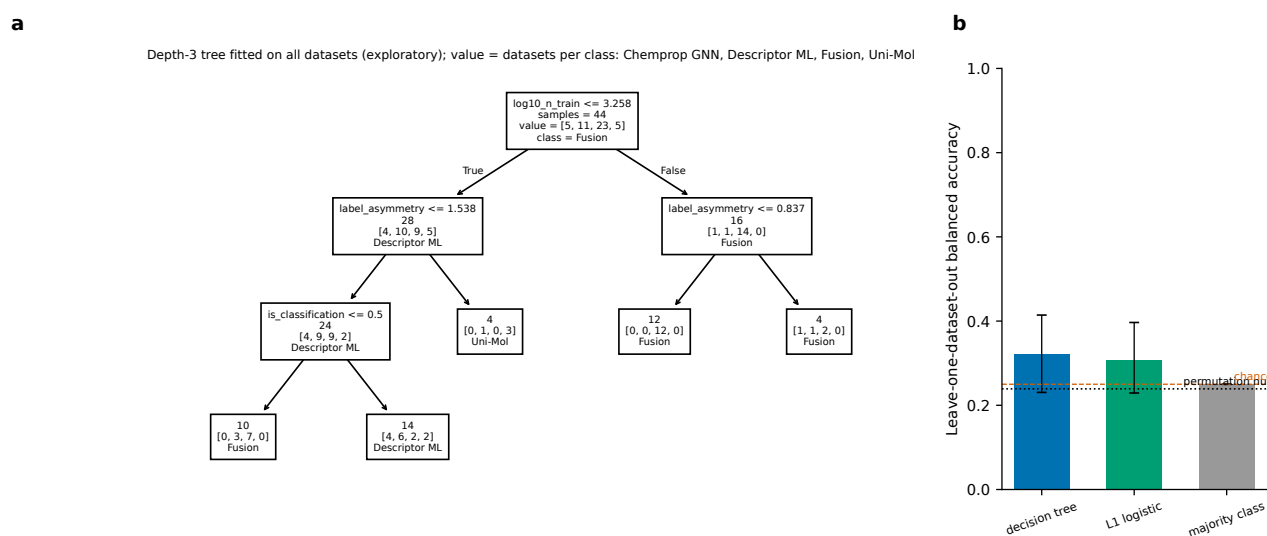


Figure S3: (a) Depth-3 decision tree over four meta-features, fitted on all datasets for display. (b) Leave-one-dataset-out balanced accuracy of the tree and an L1-penalised multinomial logistic model against the majority-class baseline (grey), the label-permutation null (dotted) and chance (dashed), with dataset-bootstrap 95% CIs. Exploratory: $n = 44$.

Supplementary Note S5: Comparison with existing automated QSAR platforms

This note gives the full comparison that main-text Section 3.12 summarizes.

Automating the QSAR model-building loop is not a new idea. The main-text Introduction distinguished QSARena from DeepChem, QSPRpred, OCHEM, ChemSAR and ADMET-AI; here we compare it with the automated QSAR systems closest to it in purpose.

QSAR Workbench [24] is the closest intellectual ancestor. Built on Pipeline Pilot, it systematically explores combinations of descriptor sets, learners and train/test splits, generating on the order of 600 models in under an hour, and it reports performance equivalent to published state-of-the-art approaches on its demonstration data. Its authors are explicit that the system has no built-in competitive model selection and that choosing among the generated models requires human judgement. That is precisely the step QSARena automates and then audits: the per-dataset winner is selected programmatically, and main-text Section 3.5 quantifies how much of the resulting apparent performance is an artifact of making that selection on held-out data. QSAR Workbench is commercial, depends on a proprietary workflow engine, and was demonstrated on two datasets (209 and ~670 compounds); QSARena is MIT-licensed, depends only on open Python packages, and is evaluated here on 44 datasets spanning 280 to 13,445 molecules.

Schrödinger AutoQSAR [25] and its successor DeepAutoQSAR occupy the same application area as this work. AutoQSAR automates model building, validation and deployment, runs on Windows, macOS and Linux through either a command line or the Maestro GUI, and is distributed as closed-source software within the commercial Schrödinger Suite. Schrödinger has published a benchmark of DeepAutoQSAR against ChemProp and DeepPurpose on the ADMET subset of the Therapeutics Data Commons [26], reporting that DeepAutoQSAR ranked among the top performers on 20 of 22 cases and clearly outperformed the comparators on 9 of them.

A direct accuracy comparison against that study is not possible, and we do not attempt one. The white paper defines “top performer” and “clearly outperforming” qualitatively rather than as a numerical criterion, does not report per-dataset metric values, and does not release per-molecule predictions, so its results cannot be re-scored under our protocol or ours under theirs. What can be compared is summarized in Table S15, and it is less than it first appears. DeepAutoQSAR’s “top performer on 20 of 22” is its standing against two named comparators, ChemProp and DeepPurpose, not a position on the TDC leaderboard; QSARena’s 22 of 22 is a rank among curated published reference values. The two numbers are not commensurable. What the two systems can be compared on is every axis that governs who can actually use them, and there they differ sharply.

Three differences deserve emphasis. On **cost and access**, the Schrödinger tools require a commercial licence and QSAR Workbench requires a Pipeline Pilot licence, whereas QSARena is free, MIT-licensed, installable with `pip`, and runnable without any installation at all in a hosted notebook (main-text Section 3.11). On **transparency**, the comparators’ benchmark evidence is a vendor white paper and a demonstration study respectively; QSARena’s is a deposited artifact trail from which every number in this paper regenerates, together with a script that fails if the text and the artifacts disagree. On **breadth**, both prior systems are single-suite; QSARena is evaluated across five.

We also note what the commercial tools plausibly do better. Neither the QSAR Workbench nor the Schrödinger products carry the coverage gaps we report in main-text Section 3.9, both are professionally supported, and DeepAutoQSAR’s benchmark reflects sustained engineering investment in a single well-tuned engine; ours is a broad but shallow sweep. A reader whose priority is a supported production pipeline on one endpoint class may reasonably prefer them; a reader who needs auditability, cross-suite breadth, or zero cost should prefer this work.

MetaQSAR [27] is a third, more recent system and the closest comparator for a regulatory audience. It is a standalone Java desktop application, free to download after registration, with fully automated and manual modes. It computes PaDEL descriptors and fingerprints, selects features by genetic algorithm, stepwise and filter methods, fits multiple linear regression, partial least squares, linear discriminant analysis, random forest and support-vector models, and ships two applicability-domain methods — a descriptor standardization approach [28] and a posterior-probability confidence

approach for classifiers — with explicit alignment to OECD and ECHA good-practice guidance. Its published evidence is four endpoint case studies, mostly environmental toxicity, in which its models matched or exceeded previously reported QSAR models on the same data, rather than public leaderboards. QSARena is complementary: where MetaQSAR is GUI-first, classical-descriptor-based and case-study-validated, QSARena is programmatic and code-free through a hosted notebook, spans deep, 3D-pretrained and fusion architectures, and is evaluated uniformly across five public benchmark suites. MetaQSAR’s regulatory framing is the one QSARena previously lacked, and main-text Section 3.13 adopts it.

Relationship to ADMET-AI, stated precisely. ADMET-AI is the reference open tool in this space. It is *not* one of the 31 models evaluated here; it appears only as a cited comparator. What QSARena integrates is the package ADMET-AI is built on: our **Chemprop v2** (D-MPNN + RDKit2D) variant concatenates a Chemprop graph representation with RDKit physicochemical descriptors, which is essentially ADMET-AI’s recipe, but it is a reimplementations through Chemprop v2 rather than ADMET-AI’s own code, package or trained weights.

That path remained incomplete in this run. The ADMET-AI-like Chemprop v2 (D-MPNN + RDKit2D) variant produced valid results on 38 of 44 datasets, so it is informative but still not a head-to-head reproduction of ADMET-AI. **We therefore cannot claim to reproduce or to beat ADMET-AI; parity is untested.** We ran no head-to-head comparison against it, and none should be inferred from the estimated ranks reported here.

Where QSARena does not lead. Published rankings are almost always each paper’s rank *at its own publication date*. Our curated reference table inherits that convention: it contains 64 rows claiming first place across 28 datasets, and **caco2_wang** alone carries four different models recorded as rank 1. Such figures cannot be compared with one another, and none of MaxQsaring, ADMETboost, ADMET-AI, Auto-ADMET or DeepAutoQSAR appears in the TDC leaderboard snapshots we captured, so no head-to-head measurement against them exists.

We therefore re-scored every reference value in the honest direction for its metric, producing one consistent ranking per dataset. This is a materially different picture from the published claims. MaxQsaring, whose paper reports first place on 19 of 22 TDC tasks [29], holds 7 first places under joint re-scoring but is top-3 on 19 of 22 with a median rank of 2 — the strongest system in the reference set, and the fairer way to state its lead. ADMETboost’s reported 18 first places [30] reduce to **zero**: it was genuinely first in 2022 and has since been overtaken on all 22 datasets, which is a caution about citing leaderboard positions as though they were durable. MapLight’s top-3 placement on 16 of 22 [5] is an author self-report we cannot re-score, and DeepAutoQSAR’s “top performer on 20 of 22” [26] is best-of-three against ChemProp and DeepPurpose in a vendor white paper rather than a leaderboard position at all — the two figures were never in competition. We also corrected three reference values that were on the wrong scale, all from a model reporting normalised targets (for example a PPBR MAE of 0.679 against a leaderboard best of 7.440).

Against that corrected reference set, QSARena’s best model per dataset reaches a median rank of 2.5 on the 22 official TDC tasks under test-based selection, and falls to a median rank of 7 under cross-validation-only selection (main-text Section 3.5). MaxQsaring’s median rank of 2 is therefore better than ours under either protocol. We make no claim to accuracy leadership. The contribution is the cross-suite benchmark and the selection-protocol decomposition; code-free access is a usability feature.

The same landscape shows that rigorous benchmarking builds credibility but does not by itself win users. The most widely used tools in this space — SwissADME [31], pkCSM [32] and ADMETlab [33] — are free, fixed-model web services that do not report on the TDC leaderboard at all, and they are popular because they are convenient. QSARena aims to offer both, and we report the two claims separately.

Table S15: Positioning of QSARena relative to representative open-source, commercial and web-based molecular property-prediction tools. Performance entries report each tool’s **own published results** and are **not** head-to-head comparisons: rows use different splits, seeds and selection protocols, and “—” indicates no public TDC leaderboard results. Where a published claim could be re-scored against the full reference set we give both, because they differ substantially: ADMETboost’s 18 reported first places become 0 once all reference values are ranked together, and MaxQsaring’s 19 become 7. None of the tools in this table appears in the TDC leaderboard snapshots we captured, so every performance entry is publication-reported. ADMET-AI’s “highest average rank” is a 2024 self-reported aggregate, not per-endpoint supremacy. QSARena reuses ADMET-AI’s Chemprop-RDKit architecture through its **Chemprop v2** (D-MPNN + RDKit2D) variant but does not integrate ADMET-AI itself, and in this run that variant produced valid results on 38 of 44 datasets, so parity with ADMET-AI remains untested. “Code-free” means usable without scripting; “re-train” means the user can fit models on their own data; “suites” counts distinct benchmark collections the tool has been evaluated on in its own publications.

Tool	Access / licence	Code-free	Retrain	Architecture breadth	Suites	Published TDC ADMET performance
QSARena (this work)	Open, MIT	Yes (Colab + CLI)	Yes	31 models, 8 families	5	Est. top-10 on 22/22 (test-selected); 16/22, median rank 7 under CV-only selection
ADMET-AI [34]	Open, MIT	Partial (web + CLI)	Yes (training scripts)	1 (Chemprop-RDKit)	1 (41 datasets)	Highest average rank on TDC ADMET leaderboard (self-reported, 2024)
MaxQsaring [29]	Open, MIT	No (Python + CLI)	Yes	Multi-fingerprint feature selection + boosting	1 (22 datasets)	Reported first on 19/22; re-scored: 7 first places, top-3 on 19/22, median rank 2
ADMETboost [30]	Open	No (Python)	Yes	1 (XGBoost on multi-fingerprint features)	1 (22 datasets)	Reported first on 18/22 (2022); re-scored: 0 first places, top-3 on 6/22
Schrödinger AutoQSAR / DeepAutoQSAR [25, 26]	Commercial	Yes (Maestro GUI + CLI)	Yes	Multi-method ML + GNN	1	Top of three on 20/22, clearly best on 9/22 — a three-way vendor comparison vs ChemProp and DeepPurpose, not a leaderboard rank
QSARtuna [35]	Open, Apache-2.0	No (Python + CLI)	Yes	Representations × learners via Optuna; + Chemprop; UQ	0 (illustrative)	—
ZairaChem [36]	Open, GPL-3.0	No (CLI)	Yes	AutoML ensemble + distillation	1 (classification only)	State of the art out of the box on TDC classification
QSPRpred [37]	Open	No (Python library)	Yes	Flexible descriptors × learners	0 (framework)	—

Tool	Access / licence	Code-free	Retrain	Architecture breadth	Suites	Published TDC ADMET performance
DeepChem [38]	Open, MIT	No (Python library)	Yes	Many graph, transformer and classical models; many featurizers	MoleculeNet loaders	—
OCHEM [39]	Web platform (registration)	Yes (web)	Yes	Many descriptor sets $\times$ ML methods; applicability domain	n/a	—
ChemSAR [40]	Free web	Yes (web)	Yes	Descriptors $\times$ scikit-learn learners	n/a	—
DeepMol [41]	Open, Apache	No (Python)	Yes	AutoML over many models	1 (22 datasets)	Competitive pipelines across 22 TDC datasets
DeepPurpose [42]	Open	No (Python)	Yes	15 encoders, 50+ models	TDC loader	Used as TDC baselines
ChemXploreML [43]	Open	Yes (desktop GUI)	Yes	Embeddings + tree ensembles	0 (physicochemical)	—
MetaQSAR [27]	Free (registration); standalone Java GUI	Yes (desktop GUI)	Yes	Classical: MLR, PLS, LDA, RF, SVM on PaDEL descriptors; 2 AD methods	0 (4 endpoint case studies)	—
QSAR Workbench [24]	Commercial (Pipeline Pilot)	Yes (protocols)	Yes	Descriptor $\times$ learner grid	0 (2 CAESAR datasets)	—
ADMET Predictor (Simulations Plus)	Commercial	Yes (GUI + CLI)	Yes	Proprietary, $\sim$ 100 endpoints	Proprietary	—
StarDrop / Auto-Modeller (Optibrium)	Commercial	Yes (GUI)	Yes	PLS, RF, SVM, GP	Proprietary	—
SwissADME [31]	Free web	Yes (web)	No (fixed)	Fixed in-house models	n/a	—
pkCSM [32]	Free web	Yes (web)	No (fixed)	Graph-based signatures	n/a	—
ADMETlab 3.0 [33]	Free web	Yes (web)	No (fixed)	Multi-task graph attention	Internal	—

Supplementary Note S6: Coverage, cost and hardware details

Model coverage and backend failures. Not every model ran on every dataset, and we report the gaps rather than silently analysing only successes (Table S4). Three sources account for nearly all of them. Task applicability: regression-only and classification-only estimators are valid on at most 22 datasets each. Guardrails and capacity: Uni-Mol V2 produced valid results on 17 of 44 datasets at the 84 M-parameter size, and its 164 M and 310 M variants produced none, so the V2 results here are a partial view of that architecture. Backend failures: the Chemprop v2 variants were substantially repaired but still did not all finish everywhere, with 38 to 42 valid datasets per variant. TabPFN produced results on 41 of the 44 datasets; on the other three its label-based results were withdrawn (Note S1). Table S13 (Note S3) shows that restricting the family comparison to the datasets on which every family produced a result leaves it unchanged. Diagnosing the larger Uni-Mol V2 variants and completing Chemprop on its remaining datasets are the first things to fix before any follow-up benchmark.

Cost. Recorded model sizes underline the accessibility argument. The models that won most of our datasets — gradient-boosted trees and the linear meta-models over them — carry no pretraining corpus at all, against published comparators MolE (~100 M parameters, pretrained on ~842 M molecules) [44] and MolGPS (~3 B parameters) [45].

Feature selection scaled sub-linearly with dataset size in this run: a log-log fit of selector time against dataset size gives a slope of 0.77 (Pearson $r = 0.58$ across 44 datasets), with a median of 295 s and a maximum of 1,003 s on `tdc_herg_karim`. This run disabled the dataset-size RandomForest fallback (main-text Section 2.5), so every dataset used the full elastic-net selector; the flatter scaling relative to our consumer-GPU run reflects 32-way parallelism rather than a change in the algorithm.

Hardware comparison. Because accessibility is a central claim of this work, we repeated the analysis on an earlier execution of the same benchmark performed on a consumer laptop GPU (NVIDIA GeForce RTX 4060 Laptop GPU, Windows, `cost_optimized` profile) and compared the two runs dataset by dataset under identical analysis code (Table S5).

Thirty-seven of the 44 datasets used byte-identical held-out partitions across the two runs, verified by split-signature hash; the other seven were re-split from random or target-quartile to scaffold splits in the A100 run and are therefore not directly comparable. On those 37 directly comparable datasets, the best single model (ensembles and fusion excluded, because the two runs built them under different protocols) changed the primary metric by a median of +0.50% — the A100 run was better on 22 datasets and the consumer-GPU run on 13. At this resolution the two hardware configurations are indistinguishable in accuracy.

The A100 bought capacity and throughput: a `full` rather than `cost-optimized` model profile, a longer Chemprop schedule (40 epochs, 3-model ensembles versus 15 epochs and single models), a longer Uni-Mol schedule with mixed precision, the addition of Uni-Mol V2 and a TensorFlow deep tabular backend, 32-way parallel feature selection, and a complete run in 112 hours rather than 155. Those changes are what produced the shifts in *which family wins* reported in main-text Section 3.2 — most notably the 3D pretrained family’s classification wins, which did not occur under the shorter schedule.

The practical implication for readers is favourable to the accessibility argument. A practitioner with a laptop GPU can expect the same accuracy from this pipeline as one with an A100; what they lose is the ability to train the most expensive backends to convergence, and the wall-clock headroom to search as widely. Neither the selection-protocol result nor the estimated placements in main-text Section 3.5 depend on datacentre hardware.

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